

Review

A review of multi-criteria decision making applications for renewable energy site selection

Meng Shao ^a, Zhixin Han ^a, Jinwei Sun ^{a, b, *}, Chengsi Xiao ^a, Shulei Zhang ^a, Yuanxu Zhao ^a^a College of Engineering, Ocean University of China, 238 Songling Road, Qingdao, 266100, China^b Institute of Oceanographic Instrumentation, Qilu University of Technology (Shandong Academy of Sciences), China

ARTICLE INFO

Article history:

Received 19 October 2019

Received in revised form

28 March 2020

Accepted 27 April 2020

Available online 3 May 2020

Keywords:

Renewable energy generation

Site selection

Multi-criteria decision making

Literature review

ABSTRACT

Multi-criteria decision making (MCDM) methods have become increasingly popular in site selection decision-making of renewable energy power plants because they consider multiple conflicting goals and decision-maker preferences. In this paper, a systematic literature review of MCDM applications for renewable energy site selection is performed, covering a total of 85 papers published from 2001 to 2018 in high-level journals (chosen from the ScienceDirect database). This paper has summarized the exclusion criteria and evaluation criteria of site selection for five energy sources. The five site selection stages, criteria selection, data normalization, criteria weighting, alternative evaluation and result validation, are revealed by content analysis. All papers are further classified by the date, energy source, journal of publication, country of author affiliation, and study area.

It is found that different energy sources emphasize different criteria; however, some similarities exist. Literature surveys and expert opinions are most common criteria selection method. Reclassification is most frequently used data normalization method. The analytic hierarchy process is popular for weighting. In alternative evaluation, geographic information systems and weighted linear combination are the most popular tools. Varying the criteria weights is most commonly used result validation method.

© 2020 Elsevier Ltd. All rights reserved.

Contents

1. Introduction	379
2. Criteria selection	380
2.1. Literature review on the criteria for location decision-making	380
2.1.1. Solar and wind on land	380
2.1.2. Offshore wind	381
2.1.3. Wave	383
2.1.4. Tidal	383
2.2. Criteria selection methods	383
2.3. Results and discussions	383
3. Criteria value normalization methods	384
3.1. Elementary methods by the nature of the criteria	384
3.1.1. Quantitative criteria	384
3.1.2. Qualitative criteria	385
3.2. Unique synthesis methods	385
3.2.1. Reclassification	385
3.2.2. Linguistic variables and fuzzy theory	386
3.2.3. Fuzzy membership function	386
3.3. Results and discussions	386

* Corresponding author. College of Engineering, Ocean University of China, 238 Songling Road, Qingdao, 266100, China.

E-mail addresses: mengya@ouc.edu.cn (M. Shao), hanzhixin@stu.ouc.edu.cn (Z. Han), chbhy03@qlu.edu.cn (J. Sun), xiaochengsi@outlook.com (C. Xiao), zhangshulei@stu.ouc.edu.cn (S. Zhang), zhaoyuanxu@stu.ouc.edu.cn (Y. Zhao).

4.	Criteria weighting	386
4.1.	Analytic hierarchy process (AHP)	387
4.2.	Analytic network process (ANP)	388
4.3.	Linguistic quantifiers	388
4.4.	Fuzzy measures	388
4.5.	Rank correlation analysis	388
4.6.	Other weighting methods	388
4.6.1.	An integrated weighting method combining a subjective method with an objective method	388
4.6.2.	Stepwise weight assessment ratio analysis (SWARA) and expert evaluations	388
4.6.3.	A method based on variable precision rough numbers	388
4.6.4.	An inference procedure using software	388
4.6.5.	The Borda count (BC)	388
4.6.6.	Revised Simos procedure	389
4.6.7.	An optimization model	389
4.6.8.	Based on the relative value of each criterion	389
4.6.9.	A simplification of the ratio estimation procedure	389
4.7.	Results and discussions	389
5.	Alternative evaluation using MCDM methods	389
5.1.	Geographic information systems (GISs)	389
5.2.	Weighted linear combination (WLC)	389
5.3.	Technique for order of preference by similarity to an ideal solution (TOPSIS)	390
5.4.	Elimination and choice translating reality (ELECTRE)	390
5.5.	Ordered weighted averaging (OWA)	391
5.6.	Data envelopment analysis (DEA)	391
5.7.	Vlsekriterijumsko KOMpromisno Rangiranje (VIKOR)	391
5.8.	Benefits, opportunities, costs and risks (BOCR) analysis	391
5.9.	Goal programming (GP)	391
5.10.	Other MCDM methods	391
5.10.1.	Weighted aggregated sum product assessment (WASPAS) and the Delphi method	391
5.10.2.	The lexicographic order	391
5.10.3.	Fuzzy inference systems (FISs)	391
5.10.4.	The intuitionistic fuzzy Choquet (IFC) operator and generalized intuitionistic fuzzy ordered geometric averaging (GIFOGA) operator	391
5.10.5.	The combination of the cloud model and the Choquet integral (CI) with respect to the λ -fuzzy measure	391
5.10.6.	The linguistic Choquet integral (LCI) operator	392
5.10.7.	A novel risk-based methodology	392
5.10.8.	Multi-Attributive Border Approximation area comparison (MABAC)	392
5.10.9.	The Borda count (BC)	392
5.10.10.	The fuzzy gamma operator	392
5.10.11.	Ideal matter-element extension (ideal ME)	392
5.10.12.	Grey cumulative prospect theory (PT)	392
5.10.13.	The Condorcet consistent rule	392
5.10.14.	The map algebra approach using logistic regression	392
5.11.	Results and discussions	392
6.	Result validation methods	393
6.1.	Comparison with existing farm locations	393
6.2.	Sensitivity analysis	393
6.3.	Comparison with other MCDM methods	393
6.4.	Other validation methods	393
6.5.	Results and discussions	393
7.	Other classifications	394
7.1.	Distribution by year of publication and renewable energy source	394
7.2.	Distribution by journals	394
7.3.	Distribution by country of author affiliation	394
7.4.	Distribution by study area	394
8.	Conclusions and limitations	394
8.1.	Literature review conclusions	394
8.1.1.	Criteria and criteria selection	394
8.1.2.	Criteria value normalization methods	396
8.1.3.	Weighting methods	398
8.1.4.	Alternative evaluation methods	398
8.1.5.	Result validation methods	398
8.1.6.	Other conclusions	398
8.2.	Limitations of this review	398
	Declaration of competing interest	398
	Acknowledgements	398
	Main criteria for each publication	399
	References	401

1. Introduction

Energy plays a leading role in global economic and industrial growth. Over 80% of the world's energy production is based on traditional fossil fuels – coal, oil and natural gas. However, over two-thirds of total greenhouse gas emissions and more than 80% of CO₂ emissions are also caused by fossil fuels [1–3]. Rising environmental concerns, depleting reserves and high energy prices have driven the search for renewable energy options. Renewable energy, including solar energy, wind energy, water energy, biomass energy, geothermal energy, marine energy and other nonfossil energy [4], are harmless or does little harm to the environment. Moreover, these types of resources are widely distributed and suitable for local development and utilization. Compared with the possible exhaustion of fossil energy, renewable energy can be recycled in nature. At present, many countries have introduced policies for the development of renewable energy, and many renewable energy applications have appeared [5–8]. Renewable energy sources are expected to be an important part of the world's future energy supply. However, there are certain problems with renewable energy development. Some of the major problems associated with renewable energy include the prioritization of resources, the selection of appropriate resources, and the determination of plant location and generation capacity [9]. Among them, the determination of plant location, also known as renewable energy site selection, is one of the critical steps in building renewable energy power plant, and it influences the electricity generating capacity and future socioeconomic benefits. As a decision-making problem, the determination of plant location has been considered by some researchers to be a single-objective decision making problem. Traditional single-criterion approaches normally aim to identify the most efficient options with regard to high potential resources or low costs. However, since 2001, an increasing number of researchers have considered the influence of other factors in addition to economic evaluation, such as environmental, technical and social aspects. Renewable energy site selection is considered a complex multi-criteria decision making (MCDM) problem, driving the applications of an increasing number of MCDM methods. For the site selection problem, which is complex, MCDM methods assist in dealing with multiple, often conflicting criteria in a structured way, allowing different preferences to be considered. Compared to single-criterion approaches, MCDM methods have the distinctive advantage of employing multiple criteria or attributes to obtain an integrated decision-making result.

As of 2018, 85 papers on the application of MCDM methods for renewable energy site selection have been published in high-level journals indexed in the ScienceDirect. There have been several reviews of the use of MCDM methods in the renewable energy field, such as fuzzy multi-criteria decision making (FMCDM) methods for energy policy-making [10], MCDM methods for the sustainability evaluation of concentrated solar power [11], MCDM methods for sustainable renewable energy development [12], applications of MCDM methods for solving energy management problems [13], MCDM methods for sustainable energy planning [14], and life cycle analysis with MCDM methods for the sustainability evaluation of renewable energy technologies [15]. To date, however, there has been no systematic review of studies on the application of MCDM methods for renewable energy site selection. This paper is the first comprehensive review of these studies. Importantly, the scope of the articles reviewed in this paper is limited to the site selection or location problem of renewable energy using MCDM methods. Furthermore, papers that use geographic information systems

(GISs) to exclude unsuitable sites without using MCDM methods to make further site selection decisions fall beyond the scope of this review.

In general, the MCDM problem for renewable energy site selection involves multiple alternatives evaluated based on multiple criteria. This process usually includes six main stages: the formulation of alternatives, criteria selection, data normalization, criteria weighting, alternative evaluation, and result validation. In a practical problem, these six stages may not be in this order or may overlap. The first stage, the formulation of alternatives, involves defining the problem and the study area, where the problem is regional suitability assessment or the ranking of the locations of one or more renewable energy sources and where the study area is determined by the researcher according to actual needs. The second stage is criteria selection. This stage is a complex and significant process of grouping and defining the exclusion criteria and evaluation criteria having a special impact on assessing the suitability of a given area for the installation of renewable energy power plants. The third stage is the collection and normalization of data on the criteria. These initial data are derived from actual numerical values or/and the opinions of the decision-maker. They are then normalized using a normalization method so that they can be used for quantitative calculations. The fourth stage is the determination of the weight coefficients of the evaluation criteria. When solving real problems, the criteria do not have the same degree of significance; thus, it is necessary to define the influence of the individual criteria using appropriate weight coefficients (weights) for them. Weighting methods are used to calculate the normalized weights of the criteria. In the fifth stage, if exclusion criteria are used, unsuitable areas are excluded from further analysis based on the exclusion criteria. Then, regional suitability assessment or the ranking of the locations is performed based on the information on the alternatives and criteria and by means of MCDM methods. The sixth stage checks the results and further analyzes them. If the validation finds that the results hold, the most suitable location for the construction of renewable energy power plants will be determined.

Specifying the research questions is a representative part of any systematic review. To study MCDM applications for renewable energy site selection, the present paper attempts to answer the following main questions: (1) Which criteria have been frequently used in site selection for solar, onshore wind, offshore wind, wave and tidal? (2) Which methods have been prevalently applied to five site selection stages of criteria selection, criteria value normalization, criteria weighting, alternative evaluation and result validation? (3) What are the statistical laws of published papers in various aspects, such as the date, energy source, journal of publication, country of author affiliation, and study area?

A systematic literature review of MCDM applications for renewable energy site selection is performed, covering a total of 85 papers published from 2001 to 2018 in high-level journals (chosen from the ScienceDirect database). In the published papers, energy sources on land include solar [16–48], onshore wind [49–77], wind and solar [78–86], while energy sources at sea include offshore wind [87–93], wave [94–96], tidal [97,98], offshore wind and wave [99,100]. Firstly, the exclusion criteria and evaluation criteria for the location decision-making of five energy sources are reviewed. Then, five site selection stages, criteria selection, data normalization, criteria weighting, alternative evaluation and result validation, are revealed by content analysis. As a research method, content analysis includes both quantitative and qualitative research strategies. Quantitative analysis gives the result in the form of frequency to

typically answer the question ‘how many’. Qualitative analysis presents data in the form of categories to enable interpretation of the text [101]. The reason for choosing content analysis is that this method could combine data which are similar to each other based on five site selection stages and convert them into forms that can be read by readers to further identify current hot spots by the frequency of the keywords [102]. At last, all papers are further classified by the date, energy source, journal of publication, country of author affiliation and study area.

Criteria and criteria selection, data normalization, criteria weighting, alternative evaluation, and result validation are presented and reviewed in Sections 2–6. Other distributions of the published papers by the year, energy source, journal of publication, country of author affiliation and study area are presented in Section 7.

2. Criteria selection

2.1. Literature review on the criteria for location decision-making

In this section, a literature review on the criteria for the location decision-making of five energy sources is presented. In the site selection process, renewable energy sources have different focuses in terms of their objectives, and here, it is necessary to summarize the criteria used according to the renewable energy sources. In general, there are two main groups of criteria that influence the suitability of sites for harnessing renewable energy: exclusion criteria and evaluation criteria. Exclusion criteria, also known as constraints or restrictive criteria in some papers, are used to identify and exclude sites that are not suitable for the siting of renewable energy farms. Evaluation criteria are used to evaluate and rank eligible sites that do not satisfy the exclusion criteria. Such criteria are mainly divided into four aspects: technical, economic, environmental and social criteria. According to the above classifications, the commonly used criteria in the published papers are summarized separately in Tables 1–14. Criteria used in each publication are shown in **Appendix**.

2.1.1. Solar and wind on land

2.1.1.1. Exclusion criteria. In this section, the exclusion criteria frequently used for solar farms and wind farms are listed. Tables 1 and 2 show that the exclusion criteria for solar farms and wind farms, respectively, are extremely similar. Based on the published papers, the most commonly used exclusion criteria for both solar

farms and wind farms are water bodies, protected and undeveloped areas, urban areas and buffers, roads, and slopes. In addition, other commonly used exclusion criteria for solar farms are solar radiation, railroads, vegetation, unsuitable land uses, and cultural heritage. Similarly, other commonly used exclusion criteria for wind farms are airports, areas of special protection for birds, transmission lines, wind speed, and settlements. As a constraint, each exclusion criterion can exclude many unsuitable areas. Therefore, considering exclusion criteria is an essential step and can reduce the research workload.

2.1.1.2. Technical criteria. Tables 3 and 4 list the most commonly used technical criteria for solar farms and wind farms, respectively. Solar radiation, slope, aspect, air temperature, and sunshine duration are frequently utilized technical criteria for solar farms site selection. Wind speed, slope, and wind power density are often found in published papers on wind farms site selection. In conclusion, resources and the efficiency of resource utilization are always required for solar farms and wind farms. Technical criteria such as solar radiation, wind speed, and slope are generally considered the most important criteria, and they are also often used as exclusion criteria, which can effectively prevent the poor performance of these criteria (e.g., wind speed values lower than 5.5 m/s) from being compensated for by the good performance of other criteria. Thus, technical subcriteria should be chosen carefully.

2.1.1.3. Economic criteria. As evaluation criteria, economic criteria are essential. The literature review found that the distance from roads, the distance from transmission lines, the distance from residential areas, and costs are the four most commonly used criteria for solar farms and wind farms site selection. In addition, other commonly used economic criteria for solar farms are the distance from railroads and the distance from electricity substations. Similarly, another commonly used economic criterion for wind farms is population density. In addition to the criteria mentioned above, other essential criteria for solar farms and wind farms are listed in Tables 5 and 6, respectively. There are few economic subcriteria, but each is very important. Most papers used the distance from transmission lines, distance from residential areas, and distance from roads. Compared with other types of evaluation criteria, the current economic criteria system is relatively simple and mature.

Table 1
Exclusion criteria for solar farms site selection on land.

Criteria	Number	Criteria	Number
Protected and undeveloped areas	16	Areas of special protection for birds	4
Water bodies	15	Rural areas and buffer	4
Urban areas and buffer	15	Settlements	4
Roads	13	Mediterranean coast	3
Slope	13	Mountains	3
Solar radiation	6	Areas of high landscape value	3
Railroads	6	Areas of high landscape value	3
Vegetation	6	Agriculture areas	3
Unsuitable land uses	6	Sand dunes	3
Cultural heritage	6	Airport	3
Forest	5	Religious sites	2
Water infrastructure, military zones and cattle trails	5	Touristic sites	2
Community interest sites	5	Population density	2
Transmission lines	4	Flood area	2
Watercourses and streams	4	Aspect	2
Archaeological sites	4	Others	13
Paleontological sites	4		

Table 2

Exclusion criteria for wind farms site selection on land.

Criteria	Number	Criteria	Number
Urban areas and buffer	14	Railroads	4
Water bodies	12	Protected landscapes	4
Roads	11	Historical sites	4
Airport	11	Cultural heritage	3
Slope	11	Vegetation	3
Protected and undeveloped areas	10	Water infrastructure	2
Areas of protection for birds	7	Community interest sites	2
Transmission lines	7	Mediterranean coast	2
Wind speed	7	Cattle trails	2
Settlements	7	Paleontological sites	2
Unsuitable land uses	5	Forest	2
Archaeological sites	5	Others	17
Military zones	4		

Table 3

Technical criteria for solar farms site selection on land.

Criteria	Number	Criteria	Number
Solar radiation	37	Cloudiness	3
Slope	22	Distance from water ways	2
Aspect	14	Distance from dams	2
Air temperature	11	Distance from underground water	2
Sunshine duration	7	Land surface temperature	2
Elevation	4	Climatic conditions	2
Humidity	4	Others	17

2.1.1.4. Environmental criteria. The main objective of renewable energy projects is to develop emission reduction projects. Thus, to assess the sustainability of renewable energy farms, the determination of environmental criteria is an essential step. From Table 7, the most commonly used environmental criteria for solar farms are land use, pollutant emission reduction benefits, and agrological capacity. As shown in Table 8, the most frequently utilized environmental criteria for wind farms are land use, pollutant emission reduction benefits, the distance to airports, visual impact, the distance to natural reserves, noise, and the impact on ecosystems. For solar farms and wind farms, none of the environmental criteria are used extremely frequently. Rather than using environmental criteria, many researchers prefer to use exclusion criteria to consider the impact of environmental factors.

Table 4

Technical criteria for wind farms site selection on land.

Criteria	Number	Criteria	Number
Wind speed	24	Area	2
Slope	15	Annual effective utilization time	2
Wind power density	12	Effect of wind gusting	2
Elevation	4	Material and grade of soil	2
Turbulence intensity	3	Real and technical availability	2
Influence of selected height of installation	3	Efficiency and reliability	2
Geographical direction	3	Power factor and capacity factor	2
Geological/topographic condition	2	Others	14

Table 5

Economic criteria for solar farms site selection on land.

Criteria	Number	Criteria	Number
Distance from roads	27	Distance to electricity substations	5
Distance from transmission lines	26	Population density	4
Distance from residential areas	20	Electricity demand	4
Costs	10	Payback period	2
Distance from railroads	6	Others	14

Table 6

Economic criteria for wind farms site selection on land.

Criteria	Number	Criteria	Number
Distance to transmission lines	20	Distance to mast	2
Distance from roads	18	Payback period	2
Distance to residential areas	15	Land price	2
Costs	9	Electricity demand	2
Population density	6	Others	14
Distance to electricity substations	3		

2.1.1.5. Social criteria. In the published papers, social criteria are frequently seen in solar farms and wind farm site selection. Tables 9 and 10 list the most commonly used social criteria for solar farms and wind farms, respectively. Public support, the impact on the local economy, and policy support are the most frequently utilized social criteria in solar farms site selection. Additionally, public support, the impact on the local economy and tourism, and policy support are the most commonly used social criteria in wind farms site selection. Out of all evaluation criteria, social subcriteria are the least frequently used, possibly due to the difficulty of obtaining operational data and the fact that in some papers, social factors have been taken into account in the exclusion criteria.

2.1.2. Offshore wind

2.1.2.1. Exclusion criteria. In the published papers, there are 9 articles on the site selection of offshore wind farms. In these papers,

Table 7

Environmental criteria for solar farms site selection on land.

Criteria	Number	Criteria	Number
Land use	11	Energy saving benefit	3
Pollutant emission reduction benefits	6	Wildlife impact	3
Agrological capacity	5	Visual impact	3
Land availability	3	Noise	2
Ecological damage	3	Others	13

Table 8

Environmental criteria for wind farms site selection on land.

Criteria	Number	Criteria	Number
Land use	11	Agrological capacity	3
Pollutant emission reduction benefits	6	Water and soil loss	3
Distance to airports	6	Vegetation	2
Visual impact	5	Intensity of natural disasters occurrence	2
Distance to natural reserves	4	Pollution	2
Noise	4	Natural disasters	2
Impact on ecosystems	3	Impact on birds	2
Distance from protected areas	3	Distance from historical sites	2
Ecological damage	3	Distance from protected landscapes	2
Energy saving benefit	3	Others	13

Table 9

Social criteria for solar farms site selection on land.

Criteria	Number	Criteria	Number
Public support	10	Local government support	1
Impact on local economy	5	Regulatory boundaries	1
Policy support	4	Resettlement and rehabilitation	1
Impact on local tourism	2	Population	1
Political risk	1	Public security	1
Effect on progress of surrounding region	1	Impact on local residential life	1

Table 10

Social criteria for wind farms site selection on land.

Criteria	Number	Criteria	Number
Public support	6	Concept conflict from entrepreneurs, policy makers, residents	1
Impact on local economy and tourism	4	Social impact	1
Policy support	3	Distance from places of interest	1
Population	2	Public security	1
Distance from tourist facilities	2	Impact on local residential life	1

Table 11

Exclusion criteria for offshore wind farms site selection.

Criteria	Number	Criteria	Number
Natural reserve area	4	Shipping routes	2
Water depth	4	Wind speed	2
Military areas	3	Others	5
Distance from shore	3		

Table 12

Technical criteria for offshore wind farms site selection.

Criteria	Number	Criteria	Number
Wind speed	5	Wave height	2
Mean wind power density	3	Others	7

exclusion criteria were used in 5 articles, which are listed in Table 11. Natural reserve areas, water depth, military areas, the distance from the shore, shipping routes, and wind speed were

used no less than 2 times. The exclusion criteria system for offshore wind farms site selection still needs to be further explored.

2.1.2.2. Evaluation criteria. In the published papers, 8 articles used technical criteria for offshore wind farms site selection. Table 12 lists the technical criteria. Wind speed, mean wind power density, wave height, and wind resources were utilized no less than 2 times. Wind resources are still very important technical criteria.

All 9 papers used economic criteria. In Table 13, the distance from the shore, water depth, and proximity to the power transmission grid were the most commonly used economic criteria. The distance from ports, investment, and provincial finance subsidies were used twice. The distance from the shore and water depth were frequently also used as exclusion criteria.

At sea, high importance is frequently attached to environmental factors, which are often considered in exclusion criteria and evaluation criteria. 6 papers used environmental criteria. From Table 14, 4 papers used environmental impact, and 2 papers used shipping density.

Offshore wind power can affect society in many ways, such as

Table 13

Economic criteria for offshore wind farms site selection.

Criteria	Number	Criteria	Number
Distance from the shore	5	Investment	2
Water depth	4	Operation and maintenance costs	2
Proximity to power transmission grid	4	Others	5
Distance from ports	2		

Table 14

Environmental criteria for offshore wind farms site selection.

Criteria	Number	Criteria	Number
Environmental impact	4	Others	3
Shipping density	2		

job creation and the living environment. Different locations of offshore wind plants have different effects on the local society. From **Appendix**, 5 papers used social criteria, and no criterion was used for multiple papers. Subcriteria such as employment and benefit to fog haze weather can usually be considered as the criteria of social benefits.

2.1.3. Wave

At present, there are 5 papers on the site selection of wave energy systems [94–96,99,100], two of which address the site selection of hybrid offshore wind and wave energy systems [99,100]. In these published papers, the exclusion and evaluation criteria for wave energy systems site selection are listed in **Appendix**, respectively. In the 3 papers that used exclusion criteria, the most frequently used are water depth, the distance from the shore, wave power density, marine protected areas and military exercise areas. In all 5 papers, the distance from the shore, the distance from ports, shipping density, water depth, wave power density, and wave height were frequently utilized evaluation criteria for wave energy systems site selection.

2.1.4. Tidal

In the published papers, there are 2 articles on tidal energy systems site selection [97,98]. The criteria of these two papers and their classification have their own characteristics. The evaluation criteria are listed in **Appendix**; there are no exclusion criteria in either paper.

2.2. Criteria selection methods

There are many different criteria for renewable energy site selection. Regarding the number of criteria, it is not the case that the more, the better, nor is it the case that the fewer, the better. Whether the criteria system is reasonable greatly affects the decision-making results. Therefore, as an important problem, the determination of criteria has been carefully considered by many researchers. In the published papers, the criteria were selected based on the following:

- literature surveys and expert opinions [16,23,25,27,35,40,52,63,81,83,86,93,97,98];
- literature surveys [19,21,32,39,64,68,77,82,87,92];
- literature surveys and national legislation [50,60];
- literature surveys, expert opinions, and the specificities of the region [36];
- literature surveys and the outcomes that the researchers aimed to achieve [47];

- the study goals, spatial scale and accessibility of a geo-referencing database [18,42];
- the spatial data and geographical coverage of the study area [44];
- scores estimated from a survey of experts, literature reviews and the data conditions of the parameters [95];
- questionnaires, workshops organized with stakeholders, expert knowledge, or a combination of all three [53];
- consultation with public and private organizations [100];
- literature surveys, national legislation, and interviews with specialist agencies [79];
- literature surveys, legislation, and data availability [89];
- literature surveys and the authors' considerations [26];
- literature surveys, the outcomes from questionnaires, and national and international guidelines [54];
- literature surveys, expert opinions, and personal experience [55];
- literature surveys and stakeholders [37]; and
- literature surveys, national and international guidelines, and data availability [67].

Most researchers selected criteria mainly based on the method of using literature surveys and expert opinions. Some researchers also considered the study goals, national and international guidelines, data availability, stakeholders, the spatial scale, national legislation, the specificities of the region, public and private organizations and interviews with specialist agencies in the process of criteria selection.

2.3. Results and discussions

The location decision-making of renewable energy plants is based on a lot of criteria due to the complexity of this issue. In this section, the most commonly used criteria for five energy sources have been listed. It is found that different energy sources focus on different site selection criteria. The criteria of solar and onshore wind are similar. Three marine energy sources, offshore wind, wave and tidal, have quite different criteria from energy sources on land. At present, the criteria systems for solar and onshore wind energy have been relatively mature. The criteria system for marine energy has not been perfected due to the lack of related research. Due to the similarity of objectives, the criteria of solar and onshore wind may be used as a reference for marine energy.

Both exclusion criteria and evaluation criteria influence the suitability of sites for harnessing renewable energy. Based on the use of these two criteria, published papers can be divided into two categories. In the first category, evaluation criteria are used alone to solve the problem of small-scale site selection using only MCDM. In the second category, exclusion criteria and evaluation criteria are used in combination to solve the problem of large-scale site selection using GIS and MCDM. As a constraint, each exclusion criterion can exclude many unsuitable areas. It is an essential step to reduce the research workload. Therefore, exclusion criteria are considered by many researchers. In fact, there are also some criteria that appear as both exclusion and evaluation criteria; the difference is that they define different ranges of criteria values. For example,

wind speed is used as an exclusion criterion when its value is below 5.5 m/s; however, it is used as an evaluation criterion when its value is above 5.5 m/s. In this way, the poor performance in the range of criteria values is solved by using exclusion criteria to reduce the compensation effect between criteria. It is found that literature surveys and expert opinions are the commonly used criteria selection method. In the papers mentioning criteria selection, almost all researchers used literature surveys to determine criteria. At present, there are still not enough studies for reference. Many researchers used table to count the frequency of criteria in the literature to determine the criteria system. Through this method, researchers and/or experts can not only determine criteria system, but also further identify the threshold and normalization method for each criterion. It is believed that the work of reviewing criteria in this paper can reduce the workload and provide reference for future researchers.

3. Criteria value normalization methods

After the criteria system of renewable energy site selection is determined, the data on the criteria are obtained; these initial data include actual numerical values or/and the opinions of the decision-makers. Most of these data on the criteria are quantitative, but some are qualitative. Additionally, due to the different directions and dimensions, these criteria are usually not commensurable. To carry out the next quantitative operations, it is necessary to normalize quantitative and qualitative criteria into directionless and dimensionless operable values. This process is called criteria value normalization or data normalization. In the current and further processing of criteria values, GIS tools and fuzzy theory play a key role. It is found that the applications of GIS tool and fuzzy theory are closely related to the selection of the criteria value normalization method. Based on GIS tool, fuzzy theory and the nature of the criteria, the criteria value normalization methods are summarized in Table 15.

3.1. Elementary methods by the nature of the criteria

3.1.1. Quantitative criteria

For the normalization of quantitative criteria values, there are two basic ideas. One involves first realizing the directionlessness and then realizing the dimensionlessness of the criteria values; the other involves realizing the directionlessness and dimensionlessness of the criteria values in one step. Below, we elaborate the first idea.

In the published papers, according to the role of the criteria, the criteria were mainly divided into benefit criteria and cost criteria. Benefit criteria positively contribute to site selection, and higher values, such as wind speed, solar radiation, and electricity demand, are preferred far more than lower values. In contrast, cost criteria negatively contribute to site selection, and lower values, such as turbulence intensity, land costs, and operation and maintenance costs, are preferred far more than higher values. The criteria systems can achieve directionlessness through a treatment that can transform cost criteria into benefit criteria. The formulas for the directionlessness of cost criteria are shown as follows [81,103]:

$$x^* = M - x \quad (1)$$

or

$$x^* = \frac{1}{x}, \quad x > 0 \quad (2)$$

where M represents the upper limit of criterion x , and x^* represents

the directionless criteria value.

In addition, the differences in dimension among various criteria render them incomparable and affect the results of the overall evaluation. Therefore, the criteria need to be dimensionless. Methods of dimensionlessness eliminate the influence of the dimension and magnitude of the original criteria through mathematical transformation. To normalize quantitative criteria, linear transformation is usually adopted. $s_i (i = 1, 2, \dots, n)$ is the i -th alternative, $x_j (j = 1, 2, \dots, m)$ is the j -th evaluation criterion, $x_{ij} (i = 1, 2, \dots, n; j = 1, 2, \dots, m)$ is the i -th sample of the j -th criterion, and x_{ij}^* are called normalized criteria values. Below, we briefly introduce five representative methods of dimensionlessness in the published papers [81,103]. Unless otherwise specified, the following criteria are assumed to be benefit criteria by default.

(1) Standardization

Standardization was used in one paper; the formula is shown as follows:

$$x_{ij}^* = \frac{x_{ij} - \bar{x}_j}{s_j} \quad (3)$$

where $\bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}$ and $s_j = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}$ are the mean value and the mean square deviation of all observed values of the j -th criteria, respectively.

(2) Extremum processing

Extremum processing was used in 6 papers; the formula is shown as follows:

$$x_{ij}^* = \frac{x_{ij} - m_j}{M_j - m_j} \quad (4)$$

where $M_j = \max_i \{x_{ij}\}$ and $m_j = \min_i \{x_{ij}\}$.

When the criteria are cost criteria, the formula is shown as follows:

$$x_{ij}^* = \frac{M_j - x_{ij}}{M_j - m_j} \quad (5)$$

This formula does not apply to constant criteria values (since the denominator is 0).

(3) Linear scaling

Linear scaling was used in 9 papers; the formula is shown as follows:

$$x_{ij}^* = \frac{x_{ij}}{x_j'} \quad (6)$$

where x_j' is a special value, and it is usually m_j , M_j or $\bar{x}_j$; and $x_j' > 0$.

(4) Vector normalization

Vector normalization was used in 6 papers; the formula is shown as follows:

$$x_{ij}^* = \frac{x_{ij}}{\sqrt{\sum_{i=1}^n x_{ij}^2}} \quad (7)$$

Table 15

Criteria value normalization methods based on GIS, fuzzy theory and the nature of criteria.

Categories 1	Categories 2	Criteria	Methods	References	
Using GIS	Using fuzzy theory	Quantitative	Fuzzy membership function	[18,19,40,57,79,91]	
		Both	Fuzzy membership function Linguistic variables& fuzzy number& Vector normalization	[26,50,51] [30,58]	
	No fuzzy theory	Quantitative	Reclassification Extremum processing	[17,36,42,44,48,52,59,60,64,66,69,75,99,100] [92]	
		Both	Reclassification	[21,22,29,32,34,37,39,41,45,47,49,54,55,67,74,76,77,96]	
			Reclassification & Extremum processing	[28,85]	
			Reclassification & Vector normalization	[25]	
	Reclassification & Linear scaling		[84]		
		Reclassification& Extremum processing & Vector normalization & Linear scaling	[53]		
	No GIS	Using fuzzy theory	Quantitative	Fuzzy membership function Linguistic variables& fuzzy number& Linear scaling	[16,87,97] [80]
			Both	Picture fuzzy set	[90]
		Interval linguistic variables		[98]	
		Linguistic variables & Standardization		[27]	
Linguistic variables& fuzzy number		[65]			
Linguistic variables& fuzzy number& Vector normalization		[88]			
Linguistic variables& fuzzy number& Linear scaling		[35]			
Linguistic variables& Intuitionistic fuzzy numbers		[93]			
		Expert scoring& Intuitionistic fuzzy numbers& Linear scaling	[70]		
No fuzzy theory		Quantitative	Extremum treatment Majorant operator	[72] [43]	
		Both	Expert scoring	[38]	
Expert scoring& Rough interval			[46]		
Expert scoring& Linear scaling	[23,63,83]				
Expert scoring& Extremum processing	[81]				
Expert scoring& Vector normalization	[82]				
Linear scaling	[86]				

(no normalization process is mentioned: [20,24,31,56,61,62,68,71,73,78,89,94,95]).

When $x_{ij} > 0$, $\sum_i (x_{ij}^*)^2 = 1$, which means that each column (that is, each criterion) has the same unit vector length, making comparisons between criteria possible. This method is mostly used with the elimination and choice translating reality (ELECTRE) method and the technique for order of preference by similarity to an ideal solution (TOPSIS).

(5) Majorant operator

The majorant operator was used in one paper; the formula is shown as follows [43]:

$$x_{ij}^* = \frac{x_{ij} - \bar{x}_j}{\max\{M_j - \bar{x}_j, \bar{x}_j - m_j\}} \quad (8)$$

where $\bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}$, $M_j = \max_i \{x_{ij}\}$ and $m_j = \min_i \{x_{ij}\}$.

3.1.2. Qualitative criteria

In the published papers, qualitative criteria are used much less frequently than quantitative criteria. A simple normalization method for qualitative criteria is expert scoring. In this scoring method, experts give the evaluation value of criteria according to their own experience and/or objective data to transform the criteria into a quantitative score. This process can realize the quantification, directionlessness and dimensionlessness of the qualitative criteria in one step. In the published papers, expert scoring was mostly

used to normalize qualitative criteria [23,38,46,63,70,81,83], and 3 papers also used it to normalize quantitative criteria [23,38,46]. Specifically, in paper [46], the scoring method was used to obtain expert evaluations with scores, and these crisp evaluations were converted into the form of rough intervals to express vagueness. Rough intervals are part of rough set theory, and they can help deal with subjective and imprecise information without many assumptions and additional adjustments. Expert scoring was used in papers that did not use GISs, and one of them combined scoring with fuzzy theory [70].

3.2. Unique synthesis methods

In the published papers, there are several normalization methods that are suitable for both quantitative and qualitative criteria, such as reclassification, linguistic variables and fuzzy theory, and fuzzy membership function. These methods have been integrated and have evolved through the basic normalization methods noted above. Fuzzy membership function is also part of fuzzy theory; here, it is listed as a normalization method because of its characteristic and different usage in the published papers. These normalization methods and their usages are described below.

3.2.1. Reclassification

In reclassification, the values of criteria are ranked and classified according to a certain grading scale. In the published papers, most papers used between 3 and 9 classes. Among them, the majority of the papers divided the criteria values into 4 or 5 classes. For

example, the evaluation scores of the criteria range from 0 to 4 points, with 0, 1, 2, 3, and 4 points indicating unsuitable, marginally suitable, suitable, moderately suitable and highly suitable, respectively. In general, the higher the class of a certain site, the higher its suitability is. As common GIS tools, reclassification methods were used in 38 papers that used GISs and did not use fuzzy theory. Additionally, reclassification was mainly used to normalize quantitative criteria. It was also used for some easily classified qualitative criteria, such as land use, aspect, and agrological capacity.

3.2.2. Linguistic variables and fuzzy theory

Many real-world decisions cannot be effectively implemented when information is incomplete, unavailable, and nonmeasurable. Fuzzy set theory was proposed by Zadeh in 1965 to solve these problems [104]. Compared with traditional set theory, fuzzy theory has more decision-making advantages when dealing with vague expressions. The application of linguistic variables was proposed by Zadeh in 1974 [105]. In an MCDM problem, decision-makers assign linguistic variables based on experience and knowledge to represent their assessment of the alternatives related to the criteria. Linguistic variables have the advantages of being easy to express and being close to the thinking mode of decision-makers. It can be represented by fuzzy numbers.

In the published papers, most authors used linguistic variables and fuzzy numbers together to normalize quantitative and qualitative criteria [30,35,58,65,80,88,93]. In these papers, the linguistic variables were mostly represented by triangular fuzzy numbers. Specifically, in paper [93], the linguistic variables were transformed into intuitionistic fuzzy numbers. Furthermore, Table 15 shows that linguistic variables and fuzzy theory are more commonly used in non-GIS environments when too many qualitative criteria exist.

In addition, there are other usages of linguistic variables or fuzzy theory in the following papers. In paper [27], quantitative criteria were transformed into linguistic variables through modified standardization, while qualitative criteria were evaluated by experts to obtain variables. This paper did not quantify the linguistic variables by transforming them into fuzzy numbers but, rather, by translating them into a linguistic scale. In paper [98], interval linguistic variables were used to normalize quantitative and qualitative criteria. In paper [90], the normalization of a picture fuzzy set was performed. In paper [70], quantitative criteria were transformed into intuitionistic fuzzy numbers, and qualitative criteria were transformed into intuitionistic fuzzy numbers by means of subjective expert scoring.

3.2.3. Fuzzy membership function

Fuzzy membership function is used to estimate the individual satisfaction degree for a special criterion for each potential site. These functions define the grade of membership of an element in a fuzzy set, taking values between zero (0 = not satisfied) and one (1 = fully satisfied). Various types of membership function can be used to describe these fuzzy sets.

In the published papers, fuzzy membership function was always used alone. Linear functions were most commonly used to mathematically express individual satisfaction degrees in 12 papers [16,18,19,26,40,50,51,57,79,87,91,97]. The formula for extremum processing is usually used in linear functions. Gaussian functions appeared in 1 paper [18], while sigmoidal functions were used in 2 papers [19,40]. In terms of usage, fuzzy membership function was used to deal with quantitative criteria in all papers, and it was used to deal with qualitative criteria in 3 papers: current land use [26,50,51]. Fuzzy membership function was used in 9 papers using GISs [18,19,26,40,50,51,57,79,91] and were used in 3 papers not using GISs [16,87,97].

3.3. Results and discussions

Simple linear transformation methods were usually adopted to normalize quantitative criteria in both GIS and non-GIS environment. They can work on their own, but in most cases, they were incorporated into other methods.

For large-scale site selection using GISs, reclassification and fuzzy membership function are the most commonly used normalization methods. As two mature GIS tools, they were considered first for large-scale site selection. Reclassification divides criteria values into multiple classes and assigns proper scores to convert floating-point values to integer values. It was mainly used to normalize quantitative criteria and some easily classified qualitative criteria. It could simplify raster data information and speed up GIS calculation. The process is easy to understand and apply because the users need only to determine different thresholds according to the values distribution over the whole area. Fuzzy membership function, as another different GIS tool, estimates individual satisfaction degree for a special criterion for each potential site using simple 0–1 scale. It was always used alone to normalize criteria values. Many functions are built into this tool, such as linear function, sigmoidal function. Users can choose corresponding function based on the direction, dimension and distribution characteristic of criteria values.

For small-scale site selection without GISs, some criteria with incomplete, unavailable and nonmeasurable values were added to consideration. Expert scoring, linguistic variables and fuzzy theory were considered first by researchers. Expert scoring, as a subjective normalization method, transforms the criteria value into a quantitative score according to the experts' evaluation based on their own experience and/or objective data. This process is extremely similar to reclassification. It realizes the quantification, directionlessness and dimensionlessness of qualitative criteria in one step. Compared with reclassification, it is more applicable to deal with qualitative and fuzzy information, while it is more dependent on the reliable judgment from experts. When too many qualitative criteria exist, linguistic variables and fuzzy theory and expert scoring have been more commonly used. Linguistic variables have the advantages of being easy to express and being close to the thinking mode of decision-makers. It can be represented by fuzzy numbers to realize the transformation from qualitative values to quantitative values for further calculation. Within the scope of the literature reviewed, this method has not yet been directly applied to normalize criteria values in articles using GISs. The reason may be that it is difficult for experts to determine linguistic variables one by one for the large amount of criteria values.

In conclusion, the determination of normalization method depends mainly on the application of GIS, fuzzy theory and criteria nature. Therefore, before determining a normalization method, the site selection scale and criteria nature need to be considered carefully and the strengths and weaknesses of methods available need to be compared.

4. Criteria weighting

All criteria have their own internal impact on the results of decision-making, and it is necessary to determine the degree of impact of each criterion. Weights are assigned to criteria to indicate their relative importance. It is necessary to obtain rational and accurate criteria weights. Three factors are usually considered to obtain such weights: the variance degree of criteria, the independence of criteria, and the subjective preference of decision-makers [44]. The determination of the weight coefficient of each criterion is the core problem in MCDM methods, and it is also a difficult problem. In general, there exist two kinds of methods: equal

weighting and rank-order weighting. Both of these methods were applied in the published papers shown in Table 16.

(1) Equal weighting In equal weighting, the criteria weights are defined as follows:

$$w_i = \frac{1}{n}, i = 1, 2, \dots, n \quad (9)$$

This method requires minimal knowledge of the decision-maker's priorities and minimal input from the decision-maker. Since Dawes and Corrigan proposed the method of equal weighting in 1974 [106], it has been popularized and applied in many decision-making problems. Furthermore, in recent years, this method has been applied in the field of renewable energy site selection [21,24,50,54,72,89,92].

(2) Rank-order weighting

Equal weighting has been criticized because it ignores the relative importance of the criteria. Accordingly, rank-order weighting was proposed, and the weights are allocated as follows:

$$w_1 \geq w_2 \geq \dots \geq w_n \geq 0 \quad (10)$$

where $\sum_{i=1}^n w_i = 1$.

In general, rank-order weighting can be divided into three categories: subjective weighting, objective weighting and combinative weighting. Subjective weighting methods rely on the decision-makers' opinion, while objective weighting methods emphasize the statistical evaluation of the given data. Each of these techniques has its own advantages and disadvantages. The potential uncertainty in expert judgments is the main disadvantage of the subjective methods, while the objective methods do not benefit from the expertise and experience of decision-makers. Accordingly, some researchers highly recommend combinative weighting methods to account for both types of subjective and objective weighting and to arrive at a single, aggregated set of criteria weights. In the published papers, almost all articles used subjective weighting, one article used objective weighting [48], and one

article used combinative weighting [93].

For the rank-order weighting process, some researchers directly specified weights based on their own experience and that of experts [24,26,49,64,84,96,100] (specified directly), while others used a mathematical weighting method to assign weights. Below, we introduce the origin, basic principles, and characteristics of several rank-order weighing methods, as well as their special use in some papers, for example, to improve the normal method or to utilize in combination with other methods.

4.1. Analytic hierarchy process (AHP)

The analytic hierarchy process (AHP) was first proposed by Saaty in 1980 [107]. The AHP has two main steps: pairwise comparison and weighted linear combination (WLC). Pairwise comparison is used to weight the different criteria, while WLC is the final stage in the AHP evaluating the alternatives. An AHP-based determination of weights includes two steps. The first step establishes a judgment matrix based on people's judgment of the relative importance of each criterion at a given level, which is scored according to a scale from 1 to 9. The second step calculates the weights based on the judgment matrix and conducts a consistency test to indicate whether the comparison made is reasonable. More than half of the papers (46 papers) chose the normal AHP as their method of weighting for the following reasons:

- The AHP is simple to understand and apply to complex issues [35];
- The process decomposes a large problem into smaller sub-problems at hierarchical levels [45];
- The process introduces the comparative importance of the criteria, showing a more reliable representation of the decision goal [95];
- The process is applicable for quantitative and qualitative criteria [83]; and
- The process checks the consistency of the decision, thus reducing the bias in the decision-making progression [17].

In addition to the normal AHP method, integrated AHP approaches are prevalent in the published papers. Six papers used the fuzzy AHP to determine the weights of the different criteria,

Table 16

The criteria weighting methods for renewable energy site selection decision-making.

Categories	Weighting methods	References	Number
Equal weighting	—	[21,24,50,54,72,89,92]	7
Rank-order weighting	Specified directly	[24,26,49,64,84,96,100]	7
	AHP	[16–18,20,22,25,26,28–32,34–36,38–42,44,45,47,50,52,54,55,58,59,63,65–67,69,75–77,80,83,85–87,91,92,95,99]	46
	ANP	[20,53,88,97]	4
	Linguistic quantifiers	[26,57,75,79]	4
	Fuzzy measures	[27,70,98]	3
	Rank correlation analysis	[81,82]	2
	An integrated combinative weighting method	[93]	1
	SWARA and expert evaluations	[23]	1
	A method based on variable precision rough numbers	[46]	1
	An inference procedure using software	[33]	1
	BC	[51]	1
	Revised Simos procedure	[37]	1
	An optimization model	[43]	1
	Based on the relative value of each criterion	[48]	1
	A simplification of the ratio estimation procedure	[74]	1

(no weighting process: [19,56,60–62,68,71,73,78,90,94]).

considering the linguistic evaluations from decision-makers [16,30,53,58,65,80]. In paper [30], the fuzzy AHP was used in conjunction with the pseudo-Delphi method, and the pseudo-Delphi method was used to obtain information from a group of experts. In addition, 2 papers used the interval type-2 fuzzy AHP by combining interval type-2 fuzzy sets and the AHP to make the decision more flexible and to greatly reduce the uncertainty [52,66].

4.2. Analytic network process (ANP)

The analytic network process (ANP) is a generalization of the AHP that deals with the interaction and dependency between different criteria, and it was first introduced by Saaty in 1996 [108]. The ANP makes it possible to model all the interactions and dependencies in a network.

In the published papers, 4 articles used the normal or integrated ANP. Paper [20] used the ANP as a weighting method because it was believed that the results generated by the ANP were closer to expert intuitions than those generated by the AHP method. Fuzzy set theory was applied with the ANP in paper [97] to account for the imprecision and vagueness in human judgments and information. Paper [53] used the decision making trial and evaluation analytic network process (DEMATEL-ANP; DANP) to determine the weight coefficients of the evaluation criteria, where pairwise comparisons of the criteria from the DEMATEL method are the input parameters for the mathematical apparatus of the ANP. Paper [88] combined the fuzzy ANP and fuzzy DEMATEL method to determine weights, where the former was used to obtain the principal eigenvector of the criteria and subcriteria in the super matrix and the latter was used to obtain the principal eigenvector of the aggregated comparison matrix due to the influences of the criteria on each other and the influences of the subcriteria on each other.

4.3. Linguistic quantifiers

Linguistic quantifiers were introduced by Yager in 1996 to obtain the order weights in the ordered weighted averaging (OWA) operator instead of criteria weights [109]. Linguistic quantifiers such as “most”, “many”, “at least half”, “some”, and “few” can be implemented by mathematical tools from fuzzy set theory, allowing the attitudes of decision-makers to be included in the decision-making process. Linguistic quantifiers provide decision-makers with flexibility based on the degree of optimism that the decision-makers would like to use.

Linguistic quantifiers were used in 4 papers [26,57,75,79]. Specifically, a regular increasing quantifier “most” was used in two papers [57,79]. In paper [75], linguistic quantifiers for different degrees of optimism were evaluated.

4.4. Fuzzy measures

The fuzzy measure concept was first proposed by Sugeno in 1974 [98]. Fuzzy measures are used to represent the degree of importance of criteria to avoid the independent assumption. The application of fuzzy measures in the MCDM field is very limited because the parameters to be determined increase exponentially as the criteria increase. Therefore, the λ -fuzzy measure, k -additive fuzzy measure and p -symmetry fuzzy measure were proposed to decrease this difficulty in practice.

In the published papers, 3 articles from the same first author (Yunna Wu) used fuzzy measures, and all of them used the λ -fuzzy measure to weight the criteria because this method is considered relatively easy to calculate and comparatively straightforward to

understand [27,70,98].

4.5. Rank correlation analysis

Rank correlation analysis was used in two papers [81,82], and it was called the method of order relation in paper [81]. Rank correlation analysis was used for weight determination by a group of experts. Using this method, experts just need to select the most important criterion in the criteria system at any time to determine the unique sequence. When there are too many criteria, rank correlation analysis is considered to avoid complicated pairwise comparison, to simplify the process and to reduce the influence of subjectivity.

4.6. Other weighting methods

4.6.1. An integrated weighting method combining a subjective method with an objective method

In paper [93], an integrated weighting method combining a subjective method with an objective method, is adopted to decide the criteria weights. In this method, to determine the criteria weights, the individual opinions of decision-makers are aggregated using the generalized intuitionistic fuzzy ordered weighted geometric interaction averaging (GIFWGIA) operator, while the entropy method was used to determine the subcriteria weights.

4.6.2. Stepwise weight assessment ratio analysis (SWARA) and expert evaluations

In paper [23], by using stepwise weight assessment ratio analysis (SWARA), the criteria weights are determined based on expert evaluations. In this method, each expert sorts the criteria from first to last, solely considering the overall outcome. The most important criterion is given the best rank, and the most trivial criterion is ranked last. The overall ranks are specified based on the mediocre value of ranks.

4.6.3. A method based on variable precision rough numbers

In paper [46], a novel method based on variable precision rough numbers was used to determine the criteria weights through a direct assessment of their importance. This method has the advantage of variable precision rough numbers in flexibly dealing with vague information. The core of this approach consists of converting the crisp importance into the form of a variable precision rough interval and then aggregating the individual variable precision rough importance weights.

4.6.4. An inference procedure using software

In paper [33], an inference procedure using IRIS software was utilized to obtain the criteria weights. This procedure does not require the user to indicate precise values. In this procedure, the expert gives initial bounds for the criteria weights and adds the cutting level and further constraints on the range of the parameters at any time. Finally, IRIS infers a “central” combination of these parameter values when the constraints are consistent.

4.6.5. The Borda count (BC)

The Borda count (BC), designed by the 18th century French mathematician Jean-Charles de Borda, is a positional voting system [110]. The Borda method is often used in MCDM processes involving multiple participants. In paper [51], the BC was used to rank and generate criteria weights. The environmental and economic factors are ranked from 2 to 0 and from 3 to 0, respectively. The scores are standardized by dividing a factor's Borda score by the total Borda score for that set of criteria.

4.6.6. Revised Simos procedure

The revised Simos procedure is an MCDM method that was proposed by Figueira and Roy in 2002 [111]. In paper [37], the revised Simos procedure was used to calculate the criteria weights by inviting stakeholders to rank a series of site suitability parameters. These ranking results provided a basis for calculating the relative normalized weights.

4.6.7. An optimization model

In paper [43], the criteria weights are calculated by an optimization model using grey cumulative prospect theory. The objective of this model is to maximize the integrated prospect value of each alternative because decision-makers prefer the alternative with the largest prospect value.

4.6.8. Based on the relative value of each criterion

In paper [48], the criteria weights were developed based on the relative value that each criterion brings to the development and operational costs. It is an objective weighting method.

4.6.9. A simplification of the ratio estimation procedure

In paper [74], a simplification of the ratio estimation procedure was used to assign weights. In this way, the most important criterion chosen first is given 100 points. The remaining criteria are weighted with respect to this criterion and are given a score of 0–100 points.

4.7. Results and discussions

Criteria weights were considered in most MCDM models. In these models, criteria weighting directly influences the decision-making result. Some researchers specified weights directly based on literature surveys, their own experience and experts' opinions, while the majority of researchers believed that mathematical weighting methods were necessary to determine criteria weights. Compared with the latter, the former seems to be more applicable when the criteria number is small and their weights is easier to distinguish.

Among mathematical weighting methods, pairwise comparison method in the AHP due to its understandability in theory, simplicity in application and robustness of its outcomes has gained a very high degree of popularity. Most researchers applied AHP to site selection in various situations, and some researchers effectively improved traditional AHP method by combining AHP with other techniques, such as fuzzy theory, pseudo-Delphi and interval type-2 fuzzy sets. These techniques can address the shortcomings of the AHP method from the perspective of improving the reliability of expert opinions. In spite of the popularity and applicability of AHP, it is not an approach applicable to all occasions. When there are excessive number of criteria and the interaction and dependency between different criteria must be dealt with, AHP will be replaced by ANP. In the same way, many methods have been introduced by researchers into renewable energy site selection to adapt to new problems. Meanwhile, these methods are waiting for more researches to validate their effectiveness and robustness.

In addition to numerous subjective weighting methods that rely on expert opinions, there is also one article that used objective information to determine weights and combined subjective method with objective methods. It is an attempt to use objective information to determine criteria weights for renewable energy site selection. In fact, the objective weighting methods have already had some applications in some fields. However, they have not yet been widely used in renewable energy site selection possibly because of their limitation on the qualitative criteria weighting.

In criteria weighting process, single technique is more common

than integrated approach, while more and more researchers have begun to use hybrid methods to make up for the shortcomings of a single method. Fuzzy theory is the one that has to be mentioned, and it plays an important role in criteria weighting. It was widely used in combination with other weighting methods to deal with the imprecision and vagueness in human judgments and information. It is foreseeable that in the future, the AHP method will still be the first choice for many researchers because of its popularity and applicability. But no single method can rank at best or worst. Every method has its own strength and weakness depending upon its application in all the consequence and objective of planning. Hybrid technique are thereby developing to tackle such situations in more studies.

5. Alternative evaluation using MCDM methods

After obtaining the criteria weights and effective criteria values, alternative preference orders are determined. In this step, if exclusion criteria are used, unsuitable areas are excluded from further analysis based on the exclusion criteria. Then, regional suitability assessment or the ranking of the locations is performed based on the aggregation of the information on alternatives and criteria by means of MCDM methods.

In the published papers, the listed MCDM methods in Table 17 were mainly applied for the alternative evaluation of renewable energy sites. Below, we continue to show the MCDM methods used in the process of alternative evaluation. The origin, basic principles, characteristics of these methods, and their special usages in some papers, for example, to improve the normal method or to utilize in combination with other methods, are introduced in this section.

5.1. Geographic information systems (GISs)

GISs are a widely used technology for the collection, storage, management, calculation, analysis, display and description of geo-referenced data, and they are valuable tools for assisting planning and decision-making in multiple contexts in which geo-referenced information plays a relevant role.

In the 85 published papers, 52 use GISs, mainly for regional suitability assessment and, in some papers, for the ranking of the locations. In addition to using GISs to visualize geospatial data, these papers mainly use the function of overlay analysis in GISs. This overlay analysis is combined with MCDM methods to form decision support tools to solve MCDM problems such as site selection and suitability modeling. Simple overlay analysis, also known as Boolean overlay operation, can be used to exclude unsuitable areas. In the published papers, almost all the articles using GISs conducted this overlay analysis. The most commonly used decision support tool is the weighted overlay tool, which combines overlay analysis with WLC, appearing in 40 papers. If the weighted overlay tool is used for suitability modeling, a higher value usually indicates a more suitable location. In addition, many other MCDM methods are processed with GISs, including TOPSIS [25,30,31,58,77], OWA [26,57,75,77,79], ELECTRE [25,33,60], the lexicographic order [60], Vlsekriterijumsko KOmpromisno Rangiranje (VIKOR) [77], Multi-Attributive Border Approximation area Comparison (MABAC) [53], the BC [89], the fuzzy gamma operator [19], and occupational repetitive actions (OCRA) [77].

5.2. Weighted linear combination (WLC)

WLC is the most commonly used method in MCDM problems for renewable energy site selection. In some papers, this method is also called the weighted sum method or weighted sum model. Their formulas are the same; thus, here, we use the term most commonly

Table 17

The reviewed MCDM methods for alternative evaluation of renewable energy sites.

Methods	References	Number
GISs	[17–19,21,22,24–26,28–34,36,37,39–41,44,45,47–55],[57–60,64,66,67,69,74–77,79,84,85,89,91,92,96,99,100]	52
WLC	[17,18,20–22,24,26,28,29,32,34,36–41,44,45,47–55,59,64,66,67,69,72,74,76,83–85,87,91,92,95,96,99,100]	46
TOPSIS	[25,30,31,35,46,58,77,80,90]	9
ELECTRE	[25,33,60,81,88,93]	6
OWA	[26,42,57,75,77,79]	6
DEA	[56,61,62,68]	4
VIKOR	[65,77]	2
BOCR analysis	[63,86]	2
GP	[78,97]	2
WASPAS& Delphi	[23]	1
Lexicographic order	[60]	1
FISs	[16]	1
IFC& GIFOGA	[70]	1
Cloud model& CI	[98]	1
LCI	[27]	1
Risk-based	[94]	1
MABAC	[53]	1
BC	[89]	1
Fuzzy gamma operator	[19]	1
Ideal ME	[82]	1
Grey cumulative prospect	[43]	1
Condorcet consistent rule	[71]	1
Map algebra approach using logistic regression	[73]	1

used in the published papers: WLC. In this method, the score of an alternative is calculated as follows:

$$S_i = \sum_{j=1}^n w_j x_{ij}, \quad i = 1, 2, \dots, m \quad (11)$$

Then, the resulting cardinal scores for each alternative can be used to rank, screen, or choose an alternative. The suitable alternative is usually the one whose score is higher.

Among the 46 papers that used WLC, 33 used WLC as the final step of the AHP to evaluate the alternatives. In other words, before alternative evaluation by the WLC method, the pairwise comparison method of the AHP is used to determine the weights of different criteria. Furthermore, the criteria weights in the remaining 13 papers were determined by other weighting methods. Specifically, papers [26,74] used WLC in a web-based GIS environment to identify preferable locations. Paper [95] used the WLC method cascaded to artificial neural network (ANN) techniques to predict the suitability index of locations. The ANN was utilized to identify and predict the interrelationship between the parameters and the feasibility index.

5.3. Technique for order of preference by similarity to an ideal solution (TOPSIS)

The TOPSIS method, which was proposed by Hwang and Yoon in 1981 [112], is one of the most well-known classical MCDM methods. The principle is simple: in a geometrical sense, the optimal choice should have the closest distance from the positive ideal solution and the longest distance from the negative ideal solution.

The TOPSIS method was used in 9 papers. In paper [31], the assessment of the alternatives according to their degree of adequacy was performed through the TOPSIS method. In paper [30,35,58,80], fuzzy TOPSIS was used to evaluate the alternatives. In paper [46], an extended TOPSIS method was used to rank the order of photovoltaic power plant sites. This method integrated the advantages of variable precision rough set theory and prospect theory. In paper [90], the TOPSIS method was extended to reasonably select

the most suitable site without a subjective supply for criteria weights. In paper [25], two different MCDM methods, TOPSIS and ELECTRE-TRI, were used to evaluate and classify suitable locations; additionally, the author established a comparison of differences and similarities between the two methods. In paper [77], the TOPSIS, OWA, OCRA and VIKOR methods were used to evaluate the wind farm location problem. These methods were chosen because each alternative can be evaluated with a GIS assessment criteria database. The Pearson correlation coefficient is used to perform the statistical comparison between the different MCDM methods.

5.4. Elimination and choice translating reality (ELECTRE)

ELECTRE was first proposed by Benayoun, Roy and Sussman in 1966 [113]. The ELECTRE method is based on the study of outranking relations and uses concordance and discordance indices to analyze the outranking relations between alternatives. The concordance and discordance indices can be viewed as measurements of satisfaction and dissatisfaction in which a decision-maker chooses one alternative over another.

ELECTRE was used in 6 papers. In paper [88], fuzzy ELECTRE was applied to calculate the decision-making matrices of subcriteria to alternatives and help effectively choose the alternative for an ideal site location solution. In paper [81], ELECTRE-II was used to evaluate seven wind/solar hybrid power stations that have been put into operation. In paper [93], the author proposed a new design of a comprehensive MCDM framework based on the ELECTRE-III method in the presence of multiple decision-makers under an incomplete information environment to handle site selection problems. In paper [33], the pessimistic version of the ELECTRE-TRI method was combined with a GIS for site selection by developing a multi-criteria model to be run IRIS, which offers decision-makers an interactive environment to make a final decision. In paper [60], the optimistic ELECTRE-TRI procedure based on preferences was applied to classify all potential sites to make a comparison with the lexicographic order. In paper [25], two different MCDM methods, ELECTRE-TRI and TOPSIS, were used to evaluate and classify suitable locations. In this paper, the optimistic ELECTRE-TRI procedure was applied.

5.5. Ordered weighted averaging (OWA)

OWA is a family or multi-criteria combination procedure that was introduced by Yager in 1988 [114]. Yager proposed the OWA function, which corresponds to a point between two extreme cases (i.e., “satisfaction of all the desired criteria” – “satisfaction of any of the desired criteria”) by combining the “and” and “or” operators; the combinations of these two operators is known as the “orand” operator. The OWA operator combines various criteria with order weights. The weights are assigned to the criteria according to their ordered position.

In the published papers, there are 6 articles that use OWA [26,42,57,75,77,79]. Specifically, in papers [57,79], the OWA algorithm was used to aggregate individual satisfaction into an overall satisfaction for each location through a fuzzy decision-making procedure. In paper [75], the author used a new aggregation operator extension (the AHP-OWA combination) to derive a wind farm land suitability index and classification using different degrees of optimism. The core of this method is that it improves upon the OWA operator by taking into account two types of weights (criterion weights and order weights). In paper [77], the OWA, OCRA, VIKOR and TOPSIS methods were used to find the most suitable location. The results of the Pearson correlation coefficient for each MCDM method are presented in the paper. In paper [42], the fuzzy linguistic OWA (FLOWA) module incorporated the fuzzy linguistic quantifier “all” into suitability analysis.

5.6. Data envelopment analysis (DEA)

The original concept of data envelopment analysis (DEA) was proposed by Farrell in 1957 [115]. DEA is an effective tool for measuring the efficiency of decision-making units (DMUs) with multiple inputs and outputs. Simply defined, this efficiency is the ratio of outputs to inputs in a system. DEA has been used as an optimization method for selecting the most suitable location.

In the published papers, DEA was used in 4 articles [56,61,62,68]. In paper [68], DEA was used to prioritize and rank cities in terms of their suitability for the construction of a wind farm. In paper [61], hierarchical DEA was used to select the best sites at two levels: the best region in each city and the priority of cities. In paper [62], the same author as that of [61] combined hierarchical DEA with fuzzy theory to improve the original method. The fuzzification of uncertain indicators makes this method more realistic. In paper [56], a hybrid method combining three techniques – simple DEA, the balanced scorecard (BSC) and Nash bargaining game theory – was used to perform ranking. Compared with simple DEA, the proposed integrated method has better accuracy and the ability to properly detect the relationship between DMUs.

5.7. Višekriterijumsko KOmpromisno Rangiranje (VIKOR)

The VIKOR method was developed by Serafim Opricovic in 1998 to solve decision problems with different criteria [116]. VIKOR ranks alternatives and determines the solution, denominating it as a “compromise” and obtaining the ideal solution.

In paper [65], the modified fuzzy VIKOR approach was used to determine the best wind energy production site. In this approach, decision-makers used linguistic variables to evaluate the ratings of alternatives with respect to criteria. In paper [77], the VIKOR, OCRA, TOPSIS and OWA methods were applied to rank the wind power plant alternatives. Moreover, the correlation coefficient was used to analyze the mutual correspondence between these MCDM methods.

5.8. Benefits, opportunities, costs and risks (BOCR) analysis

In paper [63,86], benefits, opportunities, costs and risks (BCOR) analysis was used. There are five ways (additive, probabilistic additive, subtractive, multiplicative priority powers and multiplicative) to combine the scores of each alternative under B, O, C and R to calculate the final ranking of the alternatives.

5.9. Goal programming (GP)

Goal programming (GP) was developed in the 1950s and began to receive truly substantial attention starting in the mid-1970s [78]. GP is an effective tool for the modeling, solution, and analysis of mathematical models involving conflicting goals and objectives.

In paper [78], a GP model based on a multisource multisink network was established to locate five renewable energy plants in five sites. In paper [97], fuzzy goal programming (FGP) was applied to help select dam sites and to maximize the satisfactoriness of the system under limited resources. In this process, the Delphi method was adopted to help determine the data on the potential dam sites.

5.10. Other MCDM methods

5.10.1. Weighted aggregated sum product assessment (WASPAS) and the Delphi method

In paper [23], weighted aggregated sum product assessment (WASPAS) is applied to evaluate and prioritize the alternatives (cities) for the construction of solar power plants. WASPAS was proposed in 2012 and is considered one of the most robust MCDM methods. The method is a combination of the weighted sum model (WSM) and weighted product model (WPM). Zavadskas et al. suggested WASPAS and discussed its accuracy to prove its strength relative to the separate use of the WSM and WPM.

5.10.2. The lexicographic order

In paper [60], first, a lexicographic order was used to conduct a categorical assessment of wind farm alternatives in a GIS, and then, the ELECTRE-TRI methodology was used to compare with the methods noted above. In this paper, the lexicographic order was considered simple because compared to other criteria, the wind speed criterion is much more important.

5.10.3. Fuzzy inference systems (FISs)

In paper [16], the author defined a set of if-then rules combining different adaptive neuro-fuzzy inference system (ANFIS) evaluation criteria and their weights within a fuzzy inference system (FIS) whose output is used to determine the most convenient location for constructing a solar power farm.

5.10.4. The intuitionistic fuzzy Choquet (IFC) operator and generalized intuitionistic fuzzy ordered geometric averaging (GIFOGA) operator

In paper [70], the intuitionistic fuzzy Choquet (IFC) operator and generalized intuitionistic fuzzy ordered geometric averaging (GIFOGA) operator were used to effectively process the partial compensatory problem of the performance scores of wind farm project plans. In addition to the partial compensatory condition, the author calculated the integrated performance scores under compensatory and noncompensatory conditions. Meanwhile, the rankings of the alternatives under these three conditions were shown and compared.

5.10.5. The combination of the cloud model and the Choquet integral (CI) with respect to the λ -fuzzy measure

In paper [98], based on the combination of the cloud model and

the Choquet integral (CI) with respect to the λ -fuzzy measure, an MCDM framework is established to select the optimal tidal barrage scheme. A cloud model is applied to describe the uncertainty of the criteria information. The Choquet integral with respect to the λ -fuzzy measure is introduced to measure the correlations among the criteria.

5.10.6. The linguistic Choquet integral (LCI) operator

In paper [27], a group decision-making method with the linguistic Choquet integral (LCI) was used to rank the alternative sites and to select the optimal solar thermal power plant site. The special steps of the decision framework of power plant site selection based on linguistic variables and a fuzzy measure were shown in this paper.

5.10.7. A novel risk-based methodology

Paper [94] presented a novel risk-based methodology for selecting sites for wave energy converter installation to minimize the overall economic risk. In the process, a Bayesian network (BN)-based model was developed to determine the probabilities of the influencing parameters. The BN was then extended to an influence diagram for estimating the expected utility for each decision alternative.

5.10.8. Multi-Attributive Border Approximation area comparison (MABAC)

In paper [53], MABAC was used to rank the selected viable locations for the construction of wind farms. The MABAC method was developed in 2015 by Pamucar and Cirovic at the Center for Research in the Area of Logistics Defence at the University of Defence in Belgrade [117]. The process of implementing the MABAC method is described in six steps in paper [53].

5.10.9. The Borda count (BC)

In paper [89], the BC was used to calculate the group scores of each decision alternative based on evaluation criteria selected by participants to rank offshore wind farm alternatives. Additionally, the process of site selection was implemented in a web-based participatory GIS (PGIS) environment.

5.10.10. The fuzzy gamma operator

In paper [19], the membership degrees of all criteria values were combined by a fuzzy gamma operator to obtain the map of optimal photovoltaic power plant locations. This gamma operator was the combination of a fuzzy sum and a product. The specific formula is described in the paper.

5.10.11. Ideal matter-element extension (ideal ME)

In paper [82], the matter-element extension (ME) method and its improved version, called ideal matter-element extension (ideal ME), are applied to evaluate wind/solar hybrid power station sites. The improved version is based on the principle of TOPSIS and the closeness degree (the degree of similarity).

5.10.12. Grey cumulative prospect theory (PT)

In paper [43], grey cumulative PT combined with grey theory and cumulative PT is used to study the site selection of photovoltaic power stations from the sustainability perspective. This method introduces grey theory to deal with information uncertainty.

5.10.13. The Condorcet consistent rule

In paper [71], social multi-criteria evaluation (SMCE) was implemented to solve the wind farm location problem. The means of a mathematical algorithm, known as the Condorcet consistent rule, was used to aggregate the criteria scores to obtain a final ranking of the available alternatives.

5.10.14. The map algebra approach using logistic regression

In paper [73], the map algebra approach using logistic regression was employed to establish the statistical relationship between wind energy and its spatially varying explanatory factors, and then, a map of the likelihood of wind energy development was generated. The method allowed scenario analysis to explore wind energy portfolios of different magnitudes and the impact of changes in the predictive factors.

5.11. Results and discussions

Table 17 shows that GISs and WLC are the most popular tools in alternative evaluation compared with TOPSIS, ELECTRE, OWA and DEA. WLC and GISs appeared simultaneously in most articles. The combination of these two tools was the primary choice for most researchers in the past. This may be due to the following reasons. WLC is a relatively simple method used for combinatorial analysis. It is regarded as the most straightforward, resulting in its traditional dominance as decision support tools in the GIS domain. It can be conducted through an easy-to-use tool available in GIS, weighted overlay tool, combining evaluation criteria layers with their corresponding weights to result in a single map layer. In terms of aggregating criteria weights and criteria values, the principles of OWA and WLC are extremely similar, so that OWA has also been used with GISs in many applications.

However, few researchers directly combined GISs with other MCDM methods, such as TOPSIS and ELECTRE, probably because these methods have not yet been developed into mature tools in the GIS environment due to the difficult or unclear necessity to operate. In multiple articles by the same first author (Sánchez-Lozano JM) [25,30,31,58,60], a method combining GIS with other MCDM methods such as TOPSIS and ELECTRE was introduced. The researchers divided the feasible area into a limited number of plots in GIS as alternatives, and extracted the data of these alternatives in the form of a decision matrix. The data was saved in Excel format in order to apply the MCDM method and finally the calculation results can be re-imported into the GIS to show the final suitability map. Dividing a large area into many small pieces of land as alternatives may be based on administrative divisions. Similarly, in paper [99], the authors divided the potential area into multiple alternatives in GIS environment based on several value ranges of the three criteria, and further carried out small-scale site selection. This method can realize large-scale and small-scale site selection of hybrid offshore wind and wave energy systems. It could provide a reference method for marine energy small-scale site selection.

In alternative evaluation stage, single MCDM method is commonly used. However, hybrid methods were also favored by some researchers because they can make up for the shortcoming of a single method. MCDM methods under fuzzy techniques were proposed and developed for effectively solving the MCDM problems, and numerous applications were reported in the literature. The fuzzy techniques that have been applied in published papers include fuzzy linguistic variables, fuzzy linguistic quantifier, fuzzy inference system, fuzzy Choquet integral, intuitionistic fuzzy Choquet operator, generalized intuitionistic fuzzy ordered geometric averaging operator, linguistic Choquet integral, fuzzy gamma operator and grey cumulative prospect theory.

There are many known MCDM methods such as DEA, VIKOR, BOCR, GP, ME and grey theory, and they have been widely used in decision-making problems in other fields. Some of them have been introduced into the research of renewable energy site selection. However, there are still few such studies available in the literature. Even if only a single MCDM method is used to solve the site selection problems of several renewable energy sources, there is still a

wide range of research space. Therefore, it is expected in the future research of renewable energy site selection, MCDM methods and hybrid methods will have more applications.

6. Result validation methods

In most real-world decision-making processes, there are many unavoidable degrees of uncertainty. To check the robustness of the evaluation results and to validate the feasibility of the method followed, many researchers conduct further sensitivity analysis and validation of the research results. In this paper, the result validation methods used in the published papers are divided into four categories: comparison with existing farm locations, sensitivity analysis, comparison with other MCDM methods and other validation methods. The methods that have been applied in renewable energy site selection decision-making are shown in Table 18.

6.1. Comparison with existing farm locations

In 8 papers, the existing farm locations in the research area were compared with the research results to confirm the reliability and accuracy of the results obtained.

6.2. Sensitivity analysis

Sensitivity analysis is mainly performed by varying the criteria weights, for example, giving equal weights or focusing on certain criteria. In addition, in paper [46], to examine the influence of different levels of variable precision on the final priority of sites, sensitivity analysis is conducted by changing the value of the variable precision α . In paper [93], the attitudes of decision-makers, which are varied based on different conditions or factors, are considered to reveal their influence and to check the robustness of the decision results. In paper [95], the sensitivity of the model developed for the prediction of feasibility is calculated using Sense It Software.

6.3. Comparison with other MCDM methods

Other MCDM methods are applied in the process of criteria weighting or alternative evaluation and to thus be able to make comparisons between the applied method and other MCDM

methods. The following MCDM methods are included: ELECTRE-TRI, ELECTRE-III, TOPSIS, fuzzy TOPSIS (FTOPSIS), fuzzy PT-based TOPSIS, rough TOPSIS, the AHP, the fuzzy AHP (FAHP), the fuzzy revised AHP, the WSM, the fuzzy WSM, DEA, principal component analysis (PCA) and numerical taxonomy (NT), entropy, the point method, multicriterion analysis by means of pairwise actions and criterion comparisons (MAPPAC), two interval linguistic methods, and cumulative prospect theory.

6.4. Other validation methods

In addition to the common methods mentioned above, other validation methods are used in some papers, as shown below. In papers [28,31], in the GIS field, pixel-by-pixel validation is performed to check if satisfactory results were obtained. In papers [41,81], the results obtained are compared with those of similar studies to verify the correctness of the results. In paper [38], the azimuth angle results achieved using real measurements are compared with the results based on the AHP. In paper [77], the Pearson correlation coefficient was used to evaluate the mutual correspondence between the different MCDM methods used in the paper. In paper [86], the ranking results of projects under five methods of BOCR analysis are compared, and it is found that the sequence of their rankings is exactly the same. In paper [28], field studies confirmed that the characteristics of the study area exist in real life, as shown in the map in the paper. Visual analysis and comparisons are performed between the layers with the highest value in the model and the optimal location in the study area.

6.5. Results and discussions

The sensitivity analysis and validation of result were conducted by most researchers. It is considered that the result validation is of great significance for identifying whether the model is suitable for the purpose it was designed for and checking whether satisfactory results are obtained. The method of varying the criteria weights in sensitivity analysis is the most commonly used validation method. Comparison with other MCDM methods and comparison with existing farm locations are also popular.

When researchers choose an MCDM method, it is important to consider the ability of method to take uncertainty and risk

Table 18

The result validation methods in renewable energy site selection.

Categories	Further explanation	References
Comparison with existing farm locations		[21,26,29,48,49,57,76,84]
Sensitivity analysis	Giving equal criteria weights	[30,45,50,53,55,58,85,99]
	Focusing on certain criteria	[23,27,30,39,43,45,50,51,53,55,70,72,83,88,98,99]
	Change the value of variable precision α	[46]
	Varying attitudes of decision-makers	[93]
	Sense It Software	[95]
Comparison with other MCDM methods	TOPSIS and ELECTRE TRI	[25]
	ELECTRE-TRI	[30,60]
	Fuzzy WSM, Fuzzy AHP and Fuzzy revised AHP	[58]
	AHP, Entropy, Point method; ELECTRE-III, WSM, MAPPAC, DEA	[80]
	Two interval linguistic methods	[98]
	Traditional TOPSIS, fuzzy PT-based TOPSIS, and rough TOPSIS	[46]
	PCA and NT	[61,62]
	AHP and FTOPSIS	[68]
	TOPSIS and cumulative prospect theory	[43]
Other validation methods	Pixel-by-pixel validation	[28,31]
	Comparison with results of similar studies	[41,81]
	Comparison with real measurements results	[38]
	Pearson correlation coefficient	[77]
	Compare results under five methods	[86]
	Field studies, visual analysis and comparisons	[28]

preferences into account. Various MCDM methods have their weaknesses and strengths in this ability. There is no perfect method, and researchers need to make trade-offs between various MCDM methods. In fact, choosing a most satisfactory MCDM method for a specific application is also an MCDM problem. Therefore, by comparing the results under multiple MCDM models, the weaknesses and strengths of applying different models to a specific problem can be identified, so as to provide reference for further research.

It is concluded that most of the result validations are carried out based on two site selection stages: criteria weighting and alternative evaluation. In criteria weighting stage, researchers validated results by changing criteria weight and weighting method. In alternative evaluation stage, researchers used other MCDM methods to make comparisons with the applied method and further analyzed the differences in results. Within the scope of reviewed literatures, it is found that no researcher has validated the results in criteria value normalization stage. In fact, two common normalization methods in the GIS environment, reclassification and fuzzy membership function, have great difference in principle. And the final results under these two methods are likely to be different. Therefore, it is recommended to perform result validation in data normalization stage in the future.

7. Other classifications

In addition to the above classifications presented based on the stages of renewable energy site selection, we have considered other classifications of the papers that may be of interest. In Section 7.1, the distribution of papers by year of publication and renewable energy source is shown. Section 7.2 presents the distribution of papers by the journals in which they were published. Section 7.3 shows the distribution of papers by country of author affiliation. Finally, in Section 7.4, we present the distribution of papers by study area.

7.1. Distribution by year of publication and renewable energy source

The frequency distribution and cumulative number of published articles with respect to the year of publication is shown in Fig. 1. The earliest paper appeared in 2001. The second paper did not appear until 2006. It is observed that the number of articles has increased significantly from 2008 to 2018; in particular, in the last five years, 60 papers were published, approximately 70% of the total number. The year with the most published papers was 2017, when 20 papers were published.

The distribution of all papers by renewable energy source is shown in Fig. 2. There is a large difference in the number of papers for various energy sources. Previous studies focused more on renewable energy on land than on renewable energy at sea. A total of 71 papers studied the location problem of renewable energy on land, while 14 papers studied the location problem of renewable energy at sea. Researches on solar and wind (onshore and offshore) energy account for the majority (94%) of published papers on renewable energy, while few researches on wave and tidal are published. The research on marine energy is still relatively rare. The first wave energy site selection paper appeared in 2010, and more marine energy papers did not appear until 2015. The earliest hybrid system on land appeared in 2010, while the earliest offshore hybrid system appeared in 2016. The solar-wind and wind-wave hybrid systems appeared in 9 and 2 papers, respectively. Single energy power generation often has the disadvantages of low efficiency and instability. Two kinds of energy can make up the defects for each other, and the generated electricity can be more continuous and

stable. Hybrid system was thereby utilized by some researchers to improve power generation performance by space complementarities and time complementarities. It is expected more and more hybrid system site selection studies will continue to appear.

7.2. Distribution by journals

Applications of MCDM methods in renewable energy site selection were published in 22 journals. Fig. 3 shows distribution of all papers by journals. It is found that *Renewable and Sustainable Energy Reviews* and *Renewable Energy* have published the most articles, with 20 and 19 papers, respectively, accounting for 46% of the total number. Other journals with more than 5 publications include *Energy Conversion and Management*, *Energy Policy*, *Applied Energy*, and *Energy*.

7.3. Distribution by country of author affiliation

Fig. 4 shows the distribution of all papers by the country of author affiliation. The country of first author affiliation is chosen to record rather than the nationality of the authors because it is more convenient to record. It is found that authors from 26 countries contributed to all published papers. The results show that China, with 14 papers, ranks first out of all the countries represented. Iran, Spain, the USA rank second, third and fourth with 12, 11, and 7 papers, respectively. Most authors are from Asia, Europe and North America, while all populated continents are represented.

7.4. Distribution by study area

The distribution of published papers by study area is an interesting topic. Fig. 5 shows the distribution of published papers by study area. It is found that China was the most frequently investigated study area with 14 times, followed by Iran, Spain, the USA, Turkey and the UK with 13, 12, 6, 5, 4 times, respectively. The remaining areas were studied no more than three times. The distribution by study area is similar to the distribution by country of author affiliation. Most study areas are in Asia, Europe and North America.

8. Conclusions and limitations

8.1. Literature review conclusions

In this paper, 85 studies on renewable energy site selection that applied MCDM methods were reviewed. These papers were carefully studied to identify the different criteria for renewable energy site selection and the methods used in different site selection stages based on the content analysis. In addition, all papers were classified according to the year, the renewable energy source, the journal of publication, the country of author affiliation, and the study area. The main statistical results are shown in Fig. 6 reflected by the frequency of the main elements. The conclusions are as follows:

8.1.1. Criteria and criteria selection

The location of renewable energy plants is based on many criteria due to the complexity of this issue. The most commonly used exclusion criteria and evaluation criteria of technical, economic, environmental and social aspects for five energy sources have been listed. It is found different energy sources emphasize different criteria. The criteria of solar and onshore wind energy are similar, while the criteria of renewable energy at sea are quite different from those on land. The criteria systems for solar and onshore wind energy have been relatively mature, while the criteria systems for marine energy have not been perfected due to

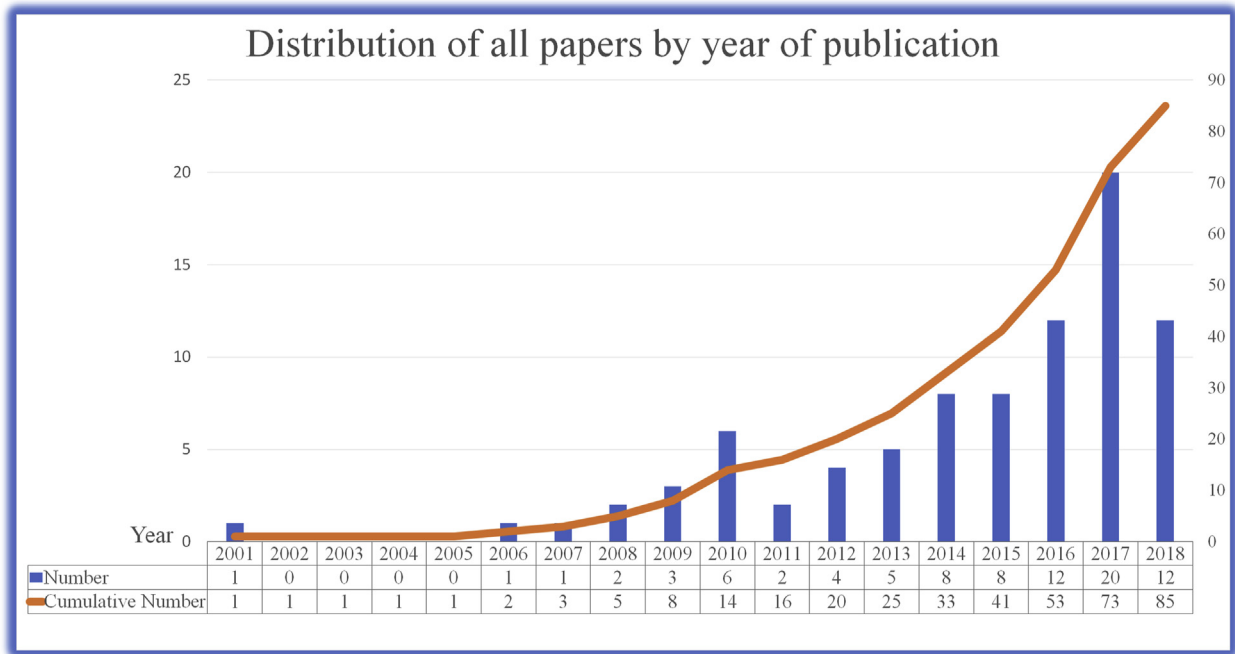


Fig. 1. Distribution of all papers by year of publication.

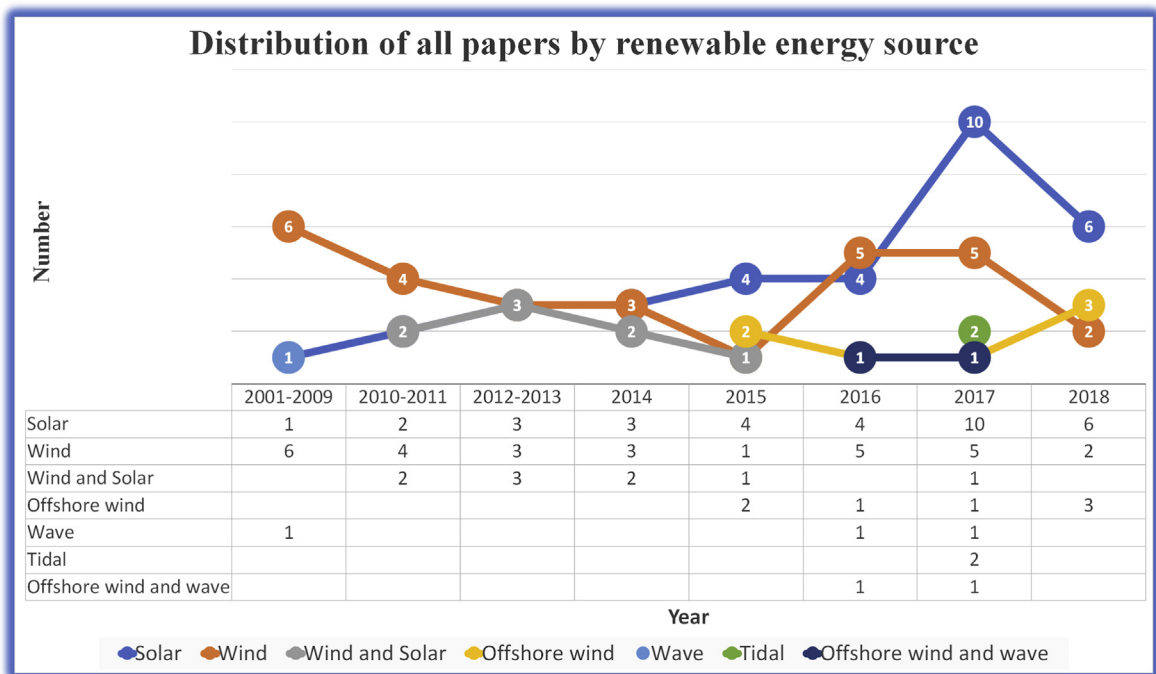


Fig. 2. Distribution of all papers by renewable energy source.

the lack of related research. Based on the use of exclusion criteria and evaluation criteria, published papers have been divided into two categories. In the first category, evaluation criteria are used alone to solve the problem of small-scale site selection using only MCDM. In the second category, exclusion criteria and evaluation criteria are used in combination to solve the problem of large-scale site selection using GIS and MCDM. The latter is more common than the former.

The criteria systems greatly influence the decision-making results; thus, criteria selection was carefully considered by many researchers. Using literature surveys and expert opinions is the most commonly used method. In the papers mentioning criteria selection process, almost all researchers used literature surveys to determine criteria. It is believed that the summary of criteria in this paper can serve as a reference for future researchers in regard to criteria selection.

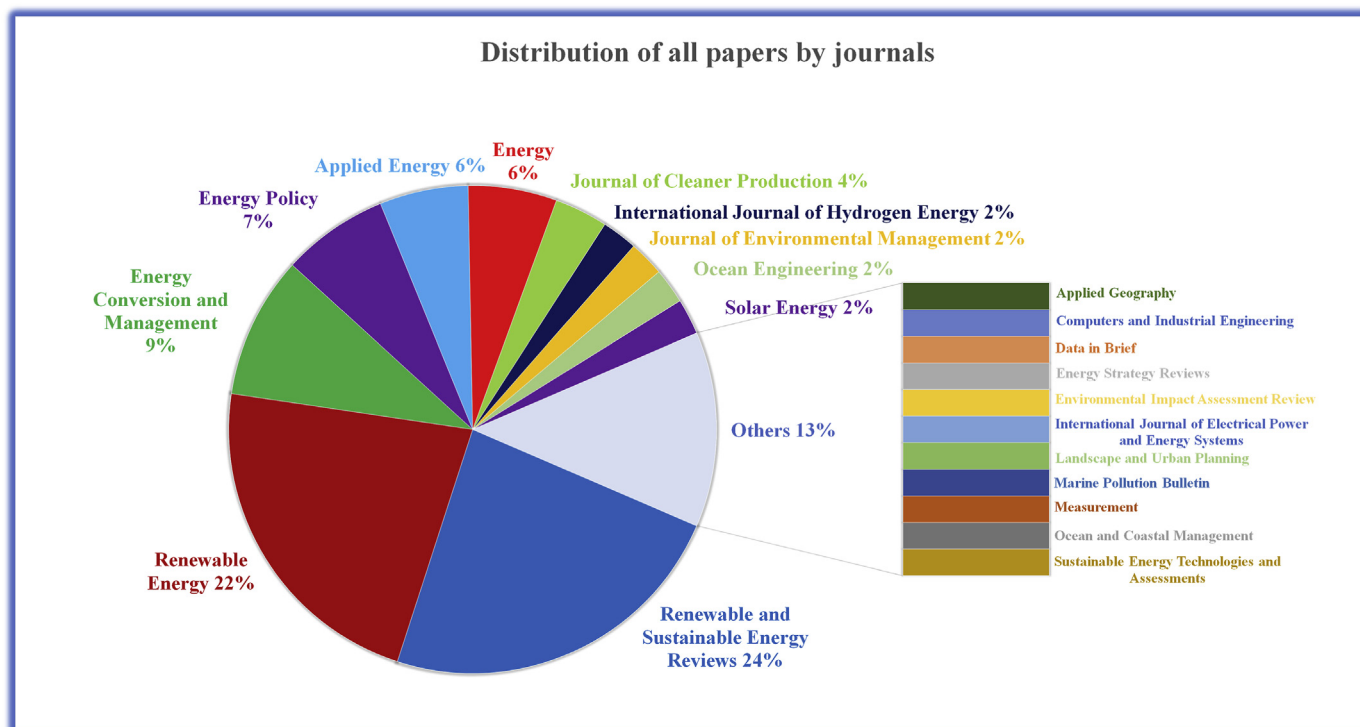


Fig. 3. Distribution of all papers by journals.

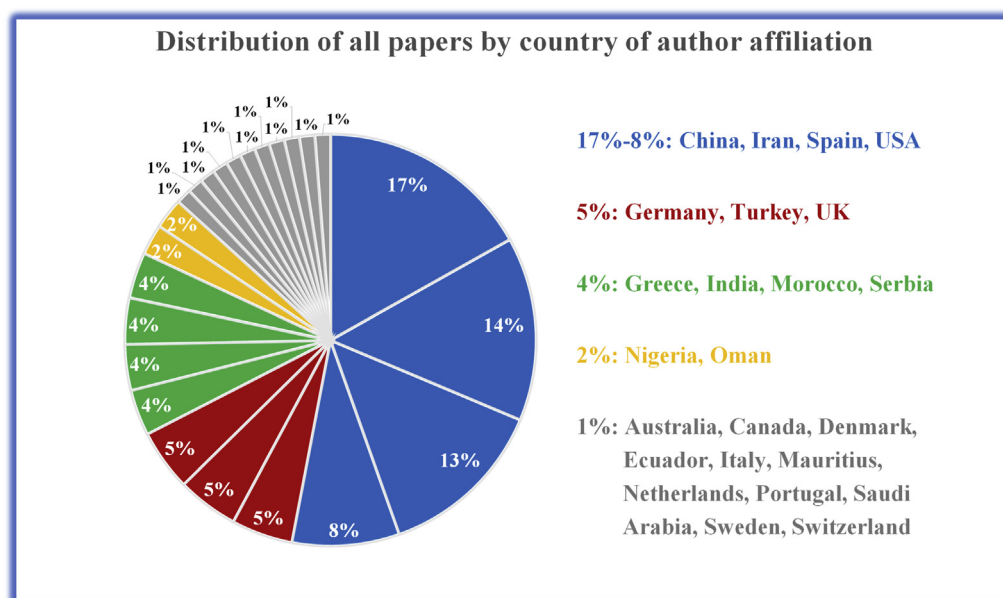


Fig. 4. Distribution of all papers by country of author affiliation.

8.1.2. Criteria value normalization methods

Normalization methods were used in many papers to solve the problem posed by the different directions and dimensions among the criteria. They have been summarized based on GIS tool, fuzzy theory and criteria nature. Simple linear transformation methods were usually incorporated into other methods to normalize quantitative criteria in both GIS and non-GIS environment. For large-scale site selection, reclassification and fuzzy membership

function, as two mature GIS tools, are most commonly used normalization methods. They can normalize quantitative criteria and some easily classified qualitative criteria. For small-scale site selection, linguistic variables and fuzzy theory and expert scoring are most commonly used normalization methods when some criteria with incomplete, unavailable and nonmeasurable values are added to consideration. Linguistic variables and fuzzy theory have the advantages of being easy to express and being close to the

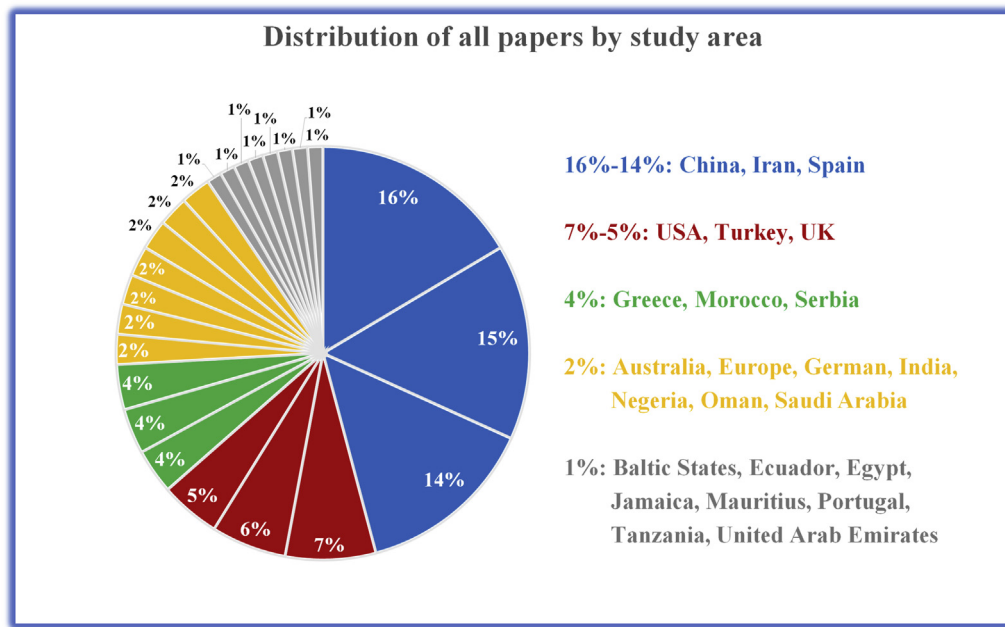


Fig. 5. Distribution of all papers by study area.

Frequency of main elements

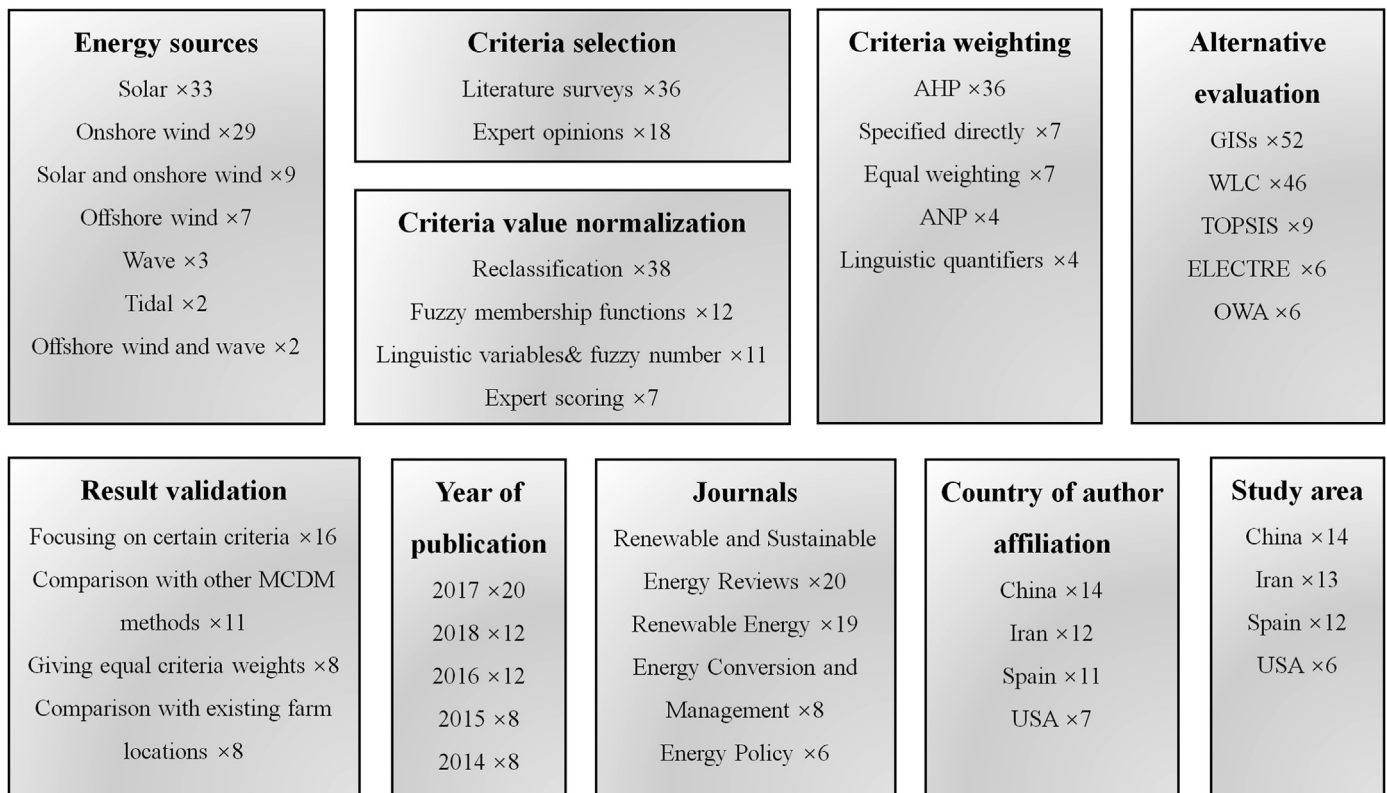


Fig. 6. Frequency of main elements.

thinking mode of decision-makers. The expert scoring process is extremely similar to reclassification. The former is more applicable to deal with qualitative and fuzzy information, and the latter is less

dependent on the reliable judgment from experts. Before determining a normalization method, the site selection scale and criteria nature need to be considered carefully and the strengths and

weaknesses of methods available need to be compared with each other.

8.1.3. Weighting methods

Criteria weighting was considered in most MCDM models because it directly influences the decision-making result. The criteria weighting methods in published papers have been listed and explained. Some researchers specified weights directly, while the majority of researchers thought that mathematical weighting methods were necessary. Compared with the latter, the former seems to be more applicable when the criteria number is small and their weights are easier to distinguish. Pairwise comparison method in the AHP has gained a very high degree of popularity due to its understandability in theory, simplicity in application and robustness of its outcomes. Some researchers have effectively improved traditional AHP by combining it with other techniques to address the shortcomings of AHP from the perspective of improving the reliability of expert opinions.

Single weighting technique is more common than integrated approach, while more and more researchers have begun to use hybrid methods to make up for the shortcomings of single method. Fuzzy theory is an important one of them. It has been widely used in combination with other weighting methods to deal with the imprecision and vagueness in human judgments and information. It is foreseeable the AHP method will still be the first choice for many researchers in the future because of its popularity and applicability. However, no single method can rank at best or worst because every method has its own strength and weakness depending upon its application in all the consequence and objectives of planning. Hybrid techniques are thereby developing to tackle such situations among more researches.

8.1.4. Alternative evaluation methods

GISs and WLC are the most popular tools in alternative evaluation compared with other techniques, such as TOPSIS, ELECTRE, OWA and DEA. The combination of WLC and GISs was the primary choice for most researchers. Few researchers directly combined GISs with other MCDM methods, such as TOPSIS and ELECTRE, probably because these methods have not yet been developed into mature tools in the GIS environment. Single MCDM method is commonly used in alternative evaluation. However, hybrid methods were also favored by some researches because they can make up for the shortcoming of a single method. MCDM methods under fuzzy techniques have been proposed and developed in numerous articles for effectively solving the MCDM problems. There are many known MCDM methods such as DEA, VIKOR, BOCR, GP, ME and grey theory that have been widely used in decision-making problems in other fields. They have been introduced into the research of renewable energy site selection; however, there are still few such studies available. Even if only a single MCDM method is used to solve the renewable energy site selection problem, there is still a wide range of research space. Therefore, it is expected that more MCDM methods and hybrid methods will be applied in study renewable energy site selection in the future.

8.1.5. Result validation methods

Result validation is of great significance for identifying whether the model is suitable for the purpose it was designed for and checking whether satisfactory results are obtained. Nearly half of the published papers adopt result validation methods to verify the decision results. These methods have been divided four categories: comparison with existing farm locations, sensitivity analysis, comparison with other MCDM methods and other validation

methods. The method of varying the criteria weights in sensitivity analysis is the most commonly used validation method. Comparison with other MCDM methods and comparison with existing farm locations are also popular. Most of the result validations are carried out based on two site selection stages: criteria weighting and alternative evaluation. It is found that no researcher has validated the results in criteria value normalization stage within the scope of reviewed literatures. Therefore, future studies can perform result validation for criteria value normalization process.

8.1.6. Other conclusions

Regarding the year of publication, relevant papers first appeared in 2001. The number of articles has increased significantly from 2008 to 2018; in particular, in the last five years, 60 papers were published, approximately 70% of the total number. Previous studies focused more on renewable energy on land with 71 papers than on renewable energy at sea with 14 papers. Research on solar and wind (onshore and offshore) energy accounts for the majority (94%) of published papers on renewable energy, while few researches on wave and tidal are published. The research on marine energy is still relatively rare. The solar-wind and wind-wave hybrid systems appeared in 9 and 2 papers, respectively. Two journals, *Renewable and Sustainable Energy Reviews* and *Renewable Energy*, have published nearly half of the papers. 28 countries contributed all papers, among which China made the largest contribution, followed by Iran and Spain. In terms of study areas, China is the most frequently investigated region, followed by Iran and Spain.

8.2. Limitations of this review

This study has some major limitations that can be considered recommendations for future studies. First, this review is focused on the use of MCDM methods for renewable energy site selection. Some papers using GISs to obtain feasible areas to further evaluate resource potential were not included in the present paper because they did not use MCDM methods. In addition, all papers were from the ScienceDirect database. Studies falling outside the scope of this database, such as academic dissertations, books, and other non-English-language research, were not addressed in our review paper. It may mean that this paper is not complete; however, it is believed that this paper comprehensively reviews and includes majority of the papers published in high-level journals. Another limitation is that all articles published in 2019 are not included in the present paper due to the limitation of time, although the articles published from 2001 to 2018 have been reviewed. Therefore, future studies can review these articles which did not mentioned in this review paper.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The research was financially supported by the National Natural Science Foundation of China (Grant No. 51609224), the Department of Science and Technology of Shandong Province (Grant Nos. 2019GGX103033, 2019GGX103015) and the Qingdao Municipal Science and Technology Bureau (Grant No. 182217jch).

Appendix. Main criteria for each publication**References Evaluation and exclusion criteria**

- [16] Evaluation: Solar radiation, Distance from roads, Distance from transmission lines, Distance from residential areas, Land availability
- [17] Evaluation: Solar radiation, Slope, Distance from water ways, Distance from dams, Distance from underground water, Distance from roads, Distance from transmission lines, Distance from residential areas, Distance from railroads
Exclusion: Water bodies, Protected and undeveloped areas, Urban areas and buffer, Roads, Railroads, Vegetation
- [18] Evaluation: Solar radiation, Distance from roads, Distance from railroads
Exclusion: Protected and undeveloped areas, Urban areas and buffer, Roads, Slope, Railroads, Vegetation, Forest, Rural areas and buffer, Infrastructure, Agriculture areas, Sand dunes, Flood area
- [19] Evaluation: Solar radiation, Slope, Distance from roads, Distance from transmission lines, Distance from residential areas, Electricity demand
Exclusion: Water bodies, Urban areas and buffer, Vegetation, Cultural heritage, Rural areas and buffer, Airport, Religious sites
- [20] Evaluation: Political risks, Technical risks, Economic risks, Time delay risks, Legal risks, Social risks
- [21] Evaluation: Slope, Aspect, Elevation, Distance from roads, Distance from transmission lines, Population density
Exclusion: Water bodies, Protected and undeveloped areas, Solar radiation, Forest
- [22] Evaluation: Solar radiation, Slope, Aspect, Cloudiness, Land surface temperature, Distance from roads, Distance from transmission lines, Distance from residential areas
Exclusion: Urban areas and buffer, Slope, Forest, Aspect
- [23] Evaluation: Solar radiation, Distance from transmission lines, Costs, Electricity demand, Payback period, Land availability, Public support, Political risk, Effect on progress of surrounding region
- [24] Evaluation: Solar radiation, Slope, Air temperature, Humidity, Distance from roads, Land use
Exclusion: Slope, Unsuitable land uses
- [25] Evaluation: Solar radiation, Slope, Aspect, Air temperature, Area, Distance from roads, Distance from transmission lines, Distance from residential areas, Distance to electricity substations, Agrological capacity
Exclusion: Protected and undeveloped areas, Urban areas and buffer, Roads, Railroads, Cultural heritage, Water infrastructure, military zones and cattle trails, Community interest sites, Watercourses and streams, Archaeological sites, Paleontological sites, Areas of special protection for birds, Mediterranean coast, Mountains, Areas of high landscape value
- [26] Evaluation: Solar radiation, Slope, Distance from transmission lines, Population density, Land use
Exclusion: Slope, Solar radiation, Unsuitable land uses, Transmission lines, Population density
- [27] Evaluation: Solar radiation, Sunshine duration, Distance from transmission lines, Costs, Pollutant emission reduction benefits, Ecological damage, Energy saving benefit, Public support, Impact on local economy, Local government support
- [28] Evaluation: Solar radiation, Slope, Aspect, Air temperature, Distance from roads, Distance from residential areas, Distance to electricity substations, Land use, Visual impact
Exclusion: Water bodies, Protected and undeveloped areas, Roads, Water infrastructure, military zones and cattle trails, Community interest sites
- [29] Evaluation: Solar radiation, Slope, Aspect, Air temperature, Sunshine duration, Humidity
Exclusion: Water bodies, Protected and undeveloped areas, Urban areas and buffer, Roads, Railroads, Vegetation, Forest, Infrastructure, Agriculture areas, Sand dunes, Airport
- [30] Evaluation: Solar radiation, Slope, Aspect, Air temperature, Area, Distance from roads, Distance from transmission lines, Distance from residential areas, Distance to electricity substations, Agrological capacity
Exclusion: Protected and undeveloped areas, Urban areas and buffer, Roads, Railroads, Cultural heritage, Water infrastructure, military zones and cattle trails, Community interest sites, Watercourses and streams, Archaeological sites, Paleontological sites, Areas of special protection for birds, Mediterranean coast, Mountains, Areas of high landscape value
- [31] Evaluation: Solar radiation, Slope, Aspect, Air temperature, Area, Distance from roads, Distance from transmission lines, Distance from residential areas, Distance to electricity substations, Agrological capacity
Exclusion: Cultural heritage, Water infrastructure, military zones and cattle trails, Community interest sites, Watercourses and streams, Archaeological sites, Paleontological sites, Areas of special protection for birds, Mediterranean coast, Mountains
- [32] Evaluation: Solar radiation, Distance from roads, Distance from transmission lines, Distance from residential areas, Population density
Exclusion: Protected and undeveloped areas, Slope, Unsuitable land uses, Settlements, Population density
- [33] Evaluation: Solar radiation, Slope, Aspect, Air temperature, Area, Distance from roads, Distance from transmission lines, Distance from residential areas, Distance to electricity substations, Agrological capacity
Exclusion: Protected and undeveloped areas, Urban areas and buffer, Roads, Railroads, Cultural heritage, Water infrastructure, Military zones and cattle trails, Community interest sites, Watercourses and streams, Archaeological sites, Paleontological sites, Areas of high landscape value
- [34] Evaluation: Slope, Distance from roads, Distance from transmission lines, Distance from residential areas, Land use
Exclusion: Water bodies, Protected and undeveloped areas, Urban areas and buffer, Rural areas and buffer
- [35] Evaluation: Climatic conditions, Distance from roads, Distance from transmission lines, Distance from residential areas, Costs, Distance from railroads, Land availability, Wildlife impact, Visual impact, Noise, Public support, Impact on local economy, Policy support, Impact on local tourism, Regulatory boundaries, Resettlement and rehabilitation
- [36] Evaluation: Solar radiation, Slope, Distance from water ways, Distance from dams, Distance from underground water, Distance from roads, Distance from transmission lines, Distance from residential areas, Distance from railroads
Exclusion: Water bodies, Urban areas and buffer, Slope, Vegetation, Transmission lines, Infrastructure
- [37] Evaluation: Solar radiation, Cloudiness, Distance from roads, Land use, Agrological capacity, Visual impact
Exclusion: Water bodies, Protected and undeveloped areas, Slope, Solar radiation, Unsuitable land uses
- [38] Evaluation: Solar radiation, Slope, Climatic conditions, Distance from transmission lines, Wildlife impact
- [39] Evaluation: Solar radiation, Slope, Aspect, Air temperature, Sunshine duration, Elevation, Humidity, Distance from roads, Distance from transmission lines
Exclusion: Water bodies, Vegetation, Cultural heritage, Settlements, Airport, Religious sites, Touristic sites
- [40] Evaluation: Solar radiation, Slope, Aspect, Sunshine duration, Elevation, Humidity, Cloudiness, Distance from roads, Distance from transmission lines, Distance from residential areas
Exclusion: Water bodies, Protected and undeveloped areas, Urban areas and buffer, Roads, Forest
- [41] Evaluation: Solar radiation, Slope, Aspect, Air temperature, Distance from roads, Distance from transmission lines, Land use
- [42] Evaluation: Solar radiation, Distance from roads
Exclusion: Water bodies, Roads, Slope, Unsuitable land uses, Rural areas and buffer, Sand dunes, Touristic sites, Flood area
- [43] Evaluation: Solar radiation, Air temperature, Elevation, Costs, Pollutant emission reduction benefits

(continued on next page)

(continued)

References	Evaluation and exclusion criteria
[44]	Evaluation: Solar radiation, Distance from roads, Distance from transmission lines, Distance from residential areas Exclusion: Water bodies, Protected and undeveloped areas, Urban areas and buffer, Slope, Solar radiation, Unsuitable land uses
[45]	Evaluation: Solar radiation, Slope, Aspect, Air temperature, Distance from transmission lines, Distance from residential areas Exclusion: Protected and undeveloped areas, Urban areas and buffer, Roads, Slope
[46]	Evaluation: Solar radiation, Slope, Aspect, Sunshine duration, Costs, Land use, Public support, Impact on local economy, Policy support
[47]	Evaluation: Solar radiation, Slope, Aspect, Land surface temperature, Distance from roads, Distance from residential areas, Land use Exclusion: Water bodies, Protected and undeveloped areas, Urban areas and buffer
[48]	Evaluation: Solar radiation, Slope, Distance from roads, Distance from transmission lines, Public support Exclusion: Water bodies, Roads, Slope, Solar radiation, Transmission lines
[49]	Evaluation: Wind speed, Slope, Land use, Vegetation Exclusion: Urban areas and buffer, Water bodies, Slope, Wind speed, Unsuitable land uses, Vegetation
[50]	Evaluation: Wind speed, Slope, Distance from roads, Land use, Distance to natural reserves, Distance from historical sites, Distance from protected landscapes, Distance from tourist facilities Exclusion: Urban areas and buffer, Roads, Airport, Slope, Wind speed, Settlements, Unsuitable land uses, Archaeological sites, Protected landscapes, Historical sites
[51]	Evaluation: Wind speed, Distance to transmission lines, Distance from roads, Population density, Land use, Impact on birds Exclusion: Water bodies, Airport
[52]	Evaluation: Wind speed, Slope, Distance to transmission lines, Distance from roads, Distance to residential areas Exclusion: Water bodies, Airport, Protected and undeveloped areas, Areas of protection for birds, Unsuitable land uses
[53]	Evaluation: Wind speed, Slope, Geographical direction, Distance to transmission lines, Distance to residential areas, Land use, Distance from protected areas, Population, Distance from tourist facilities Exclusion: Urban areas and buffer, Roads, Airport, Slope, Protected and undeveloped areas, Transmission lines, Wind speed, Unsuitable land uses, Military zones
[54]	Evaluation: Distance from roads, Agrological capacity Exclusion: Water bodies, Roads, Slope, Transmission lines, Settlements, Historical sites, Forest
[55]	Evaluation: Wind speed, Slope, Distance to transmission lines, Distance from roads, Land price, Electricity demand, Land use, Visual impact Exclusion: Water bodies, Roads, Airport, Slope, Protected and undeveloped areas, Wind speed, Settlements, Archaeological sites, Forest
[56]	Evaluation: Wind speed, Costs, Pollution
[57]	Evaluation: Wind power density, Distance to residential areas, Distance to airports, Distance to natural reserves Exclusion: Urban areas and buffer, Water bodies, Airport, Areas of protection for birds, Settlements
[58]	Evaluation: Wind speed, Slope, Area, Distance to transmission lines, Distance from roads, Distance to residential areas, Distance to electricity substations, Distance to mast, Agrological capacity Exclusion: Urban areas and buffer, Water bodies, Roads, Protected and undeveloped areas, Areas of protection for birds, Archaeological sites, Military zones, Railroads, Protected landscapes, Cultural heritage, Water infrastructure, Community interest sites, Mediterranean coast, Cattle trails, Paleontological sites
[59]	Evaluation: Wind speed, Distance to transmission lines, Distance from roads, Distance to residential areas Exclusion: Roads, Airport, Transmission lines, Wind speed, Settlements
[60]	Evaluation: Wind speed, Slope, Area, Distance to transmission lines, Distance from roads, Distance to residential areas, Distance to electricity substations, Distance to mast, Distance to airports, Agrological capacity Exclusion: Urban areas and buffer, Water bodies, Roads, Protected and undeveloped areas, Areas of protection for birds, Archaeological sites, Military zones, Railroads, Protected landscapes, Cultural heritage, Water infrastructure, Community interest sites, Mediterranean coast, Cattle trails, Paleontological sites
[61]	Evaluation: Wind speed, Slope, Elevation, Geographical direction, Material and grade of soil, Costs, Population density, Intensity of natural disasters occurrence
[62]	Evaluation: Wind speed, Slope, Elevation, Geographical direction, Material and grade of soil, Population density, Intensity of natural disasters occurrence
[63]	Evaluation: Wind speed, Wind power density, Influence of selected height of installation, Effect of wind gusting, Real and technical availability, Efficiency and reliability, Power factor and capacity factor, Distance to transmission lines, Costs, Policy support, Concept conflict from entrepreneurs, policy makers, residents
[64]	Evaluation: Wind speed, Distance to transmission lines, Distance from roads Exclusion: Urban areas and buffer, Water bodies, Roads, Slope, Protected and undeveloped areas, Transmission lines, Railroads, Historical sites, Cultural heritage
[65]	Evaluation: Visual impact, Noise, Impact on ecosystems, Public support
[66]	Evaluation: Wind speed, Slope, Elevation, Distance to transmission lines, Distance from roads, Distance to residential areas Exclusion: Urban areas and buffer, Water bodies, Roads, Airport, Slope, Protected and undeveloped areas, Areas of protection for birds, Transmission lines, Wind speed, Unsuitable land uses
[67]	Evaluation: Wind speed, Slope, Distance from roads, Distance to residential areas, Land use
[68]	Evaluation: Wind power density, Geological/topographic condition, Distance to transmission lines, Population density, Land price, Natural disasters
[69]	Evaluation: Slope, Elevation, Distance to residential areas, Visual impact, Distance from protected areas Exclusion: Urban areas and buffer, Airport, Protected and undeveloped areas, Military zones
[70]	Evaluation: Wind speed, Wind power density, Influence of selected height of installation, Annual effective utilization time, Effect of wind gusting, Costs, Payback period, Pollutant emission reduction benefits, Noise, Ecological damage, Energy saving benefit, Water and soil loss
[71]	Evaluation: Pollutant emission reduction benefits, Visual impact, Noise, Impact on ecosystems, Water and soil loss, Impact on local economy and tourism, Social impact Exclusion: Urban areas and buffer
[72]	Evaluation: Distance to residential areas, Distance from protected areas
[73]	Evaluation: Wind power density, Distance to transmission lines, Distance from roads, Population density, Distance to airports
[74]	Evaluation: Visual impact, Impact on ecosystems, Impact on local economy and tourism
[75]	Evaluation: Wind power density, Turbulence intensity, Distance from roads Exclusion: Urban areas and buffer, Slope
[76]	Evaluation: Wind speed, Slope, Distance to transmission lines, Distance from roads, Distance to residential areas, Land use, Distance to natural reserves, Distance from protected landscapes, Distance from places of interest Exclusion: Urban areas and buffer, Water bodies, Roads, Slope, Areas of protection for birds, Transmission lines, Railroads, Vegetation
[77]	Evaluation: Wind speed, Slope, Distance to transmission lines, Distance from roads, Distance to residential areas, Distance to electricity substations, Land use, Vegetation Exclusion: Urban areas and buffer, Airport, Protected and undeveloped areas, Archaeological sites, Vegetation
[78]	Evaluation: Costs, Pollutant emission reduction benefits, Public support, Impact on local economy
[79]	Evaluation: Solar radiation, Slope, Wind power density, Distance from transmission lines, Distance from residential areas, Distance to airports, Distance to natural reserves, Impact on birds

(continued)

References	Evaluation and exclusion criteria
	Exclusion: Water bodies, Protected and undeveloped areas, Urban areas and buffer, Roads, Slope, Solar radiation, Transmission lines, Areas of special protection for birds, Settlements, Airport
[80]	Evaluation: Solar radiation, Wind power density, Distance from transmission lines, Costs, Natural disasters, Population
[81]	Evaluation: Solar radiation, Wind power density, Turbulence intensity, Electricity demand, Costs, Pollutant emission reduction benefits, Ecological damage, Energy saving benefit, Public support
[82]	Evaluation: Solar radiation, Wind power density, Sunshine duration, Wind speed, Turbulence intensity, Geological/topographic condition, Distance from roads, Costs, Distance from railroads, Electricity demand, Land use, Pollutant emission reduction benefits, Pollution, Public support
[83]	Evaluation: Solar radiation, Wind speed, Sunshine duration, Wind power density, Annual effective utilization time, Payback period, Pollutant emission reduction benefits, Ecological damage, Energy saving benefit, Noise, Water and soil loss, Public support, Impact on local economy, Policy support, Impact on local tourism, Public security, Impact on local residential life Exclusion: Water bodies
[84]	Evaluation: Solar radiation, Wind speed, Distance from roads, Distance from transmission lines, Distance from residential areas, Population density, Land use
[85]	Evaluation: Solar radiation, Wind speed, Distance from roads, Distance from transmission lines, Distance from residential areas, Distance from railroads, Wildlife impact, Distance from historical sites Exclusion: Solar radiation, Agriculture areas, Aspect, Slope, Settlements, Protected landscapes, Historical sites
[86]	Evaluation: Solar radiation, Wind speed, Wind power density, Influence of selected height of installation, Real and technical availability, Efficiency and reliability, Power factor and capacity factor, Costs, Land use, Public support, Policy support
[87]	Evaluation: Wind speed, Mean wind power density, Wave height, Duration of mean wind speed, Nameplate capacity, Distance from the shore, Water depth, Distance from anchorage, Distance from fairway, Distance from fishing area
[88]	Evaluation: Wind speed, Wave height, Multiple resource use conflict, Physical feasibility, Technical feasibility and adequate wind regime, Alternative sites review, Sufficient study times, Proximity to power transmission grid, Proximity to area of electric demand, Commercial feasibility, Economic externalities, Local economic benefit, Ratio of local benefits to impacts, Ratio of power generation to impacts, Cost-benefit analysis, Environmental impact, Noise impact (construction and operation), Proximity to migratory paths for birds, Proximity to migratory paths for marine life, Proximity to facilities for construction, Impact of navigation, aviation, Community acceptance, Nearby shoreline sparsely inhabited, Mitigation of adverse local impacts, Criteria defined by federal, state, local agencies and public, Federal and/or state offshore development regulatory program in place, Adequate consideration of alternative sites, Aesthetic considerations, Collaborative process, Collaborative sitting
[89]	Evaluation: Distance from the shore, Proximity to power transmission grid, Sport fishery effort, Commercial fishery effort, Population density, Bird habitat, Fish habitat, Distance from navigable waterways
[90]	Evaluation: Construction and maintenance condition, Supporting conditions onshore, Economic, Environmental impact, Social benefit
[91]	Evaluation: Mean wind power density, Soil properties, Distance from the shore, Water depth, Proximity to power transmission grid Exclusion: Natural reserve area, Water depth, Military areas, Distance from shore, Shipping routes, Cables and tunnels, Oil and gas areas
[92]	Evaluation: Reliability, Capacity factor, Correlation, Investment, Operation and maintenance costs Exclusion: Water depth, Exclusive economic zones, Maximum radius from onshore assets, Environmental restriction zones, Heritage restriction zones, Urban restriction zones, Geological restriction zones
[93]	Evaluation: Wind speed, Mean wind power density, Effective wind hours, Wind shear, Turbulence, Meteorological conditions, Marine conditions, Undersea geological condition, Beach width, Traffic condition, Electrical transmission and distribution system, Distance to load center, Distance from the shore, Water depth, Investment, Operation and maintenance costs, Total project payback period, Expected benefit to cost ratio, Provincial finance subsidies, Environmental impact, Marine life coordination, Bird coordination, Employment, Benefit to fog haze weather Exclusion: Natural reserve area, Military areas, Harbor and channel area, Engineering infrastructures
[94]	Evaluation: Distance from shore, Sea environment, Energy flux, Wave breaking, Cost
[95]	Evaluation: Shipping density, Water depth, Wave height, Coastal erosion, Tourism potential, Wind speed, Distance between two waves, No. of waves, Turbulence, Water quality
[96]	Evaluation: Distance from shore, Distance from ports, Waves height, Sea bottom geology, Distance to power grid, Waves period, Waves power Exclusion: Water depth, Distance from shore, Marine protected areas, Military exercise areas, Underwater cables, Wave shadow areas, Harbor entrances and navigation channels
[97]	Evaluation: Business drive: Tidal data, Site advantage, Functions of tidal energy conversion, Investment return, Complementary functions; Potential opportunity: Functional diversification, Geographically balanced energy supply, Technological improvement, Supporting regulations, Financial supports; Overall costs: Levelized cost of energy, Operations and maintenance costs, Manufacturing costs, Deployment costs, Transmission and grid costs, Switching opportunity costs; Uncertainty and conflict: Cultural conflict, Technological uncertainty, Financial uncertainty, Legal uncertainty
[98]	Evaluation: Hydrodynamic impact: Tidal range reduction, Water quality decrease, Flow velocity reduction, Tidal cycle alteration; Biological impact: Fish decline, Bird loss, Algae decline, Littoral vegetation decline; Morphological impact: Sediment pattern change, Intertidal habits reduction, Coastal erosion increase, Land drainage blocked
[99]	Evaluation: Wind speed, Distance from the shore, Water depth, Proximity to power transmission grid, Distance from ports, Population served, Shipping density, Wave power density Exclusion: Natural reserve area, Water depth, Military areas, Distance from shore, Wind speed, Areas to be licensed for exploration and exploitation of hydrocarbons, Areas where offshore renewable energy projects are planned to be or have been installed, Wave power density, Marine protected areas, Military exercise areas
[100]	Evaluation: Wind speed, Power extraction, Met-ocean conditions, Weather windows, Distance from ports, Environmental impact, Shipping density, Distance from shore, Wave power density Exclusion: Natural reserve area, Water depth, Distance from shore, Shipping routes, Wind speed, Distance to operation and maintenance port, Distance to construction port, Wave power density, Natura 2000 sites, Shipping density category

References

- [1] IEA (International Energy Agency), World Energy Balance 2018: Overview, IEA Webstore, 2017, p. 13.
- [2] International Energy Agency (IEA), CO₂ emissions from fuel combustion 2018 highlights, Int. Energy Agency 1 (2018) 164.
- [3] J.L. Aleixandre-Tudó, L. Castelló-Cogollos, J.L. Aleixandre, R. Aleixandre-Benavent, Renewable energies: worldwide trends in research, funding and international collaboration, Renew. Energy 139 (2019) 268–278.
- [4] Standing Committee of the National People's Congress, Renewable Energy Law of the People's Republic of China, 2009.
- [5] J. Liu, China's renewable energy law and policy: a critical review, Renew. Sustain. Energy Rev. 99 (2019) 212–219.
- [6] D. Fouquet, Policy instruments for renewable energy - from a European perspective, Renew. Energy 49 (2013) 15–18.
- [7] S. Kim, H. Lee, H. Kim, D.H. Jang, H.J. Kim, J. Hur, et al., Improvement in policy and proactive interconnection procedure for renewable energy expansion in South Korea, Renew. Sustain. Energy Rev. 98 (2018) 150–162.
- [8] A. Tavana, A. Emami Javid, E. Houshfar, A. Mahmoudzadeh Andwari, M. Ashjaee, S. Shoaee, et al., Toward renewable and sustainable energies perspective in Iran, Renew. Energy 139 (2019) 1194–1216.
- [9] A. Aktas, M. Kabak, A model proposal for locating wind turbines, Procedia Comput. Sci. 102 (2016) 426–433.
- [10] İ. Kaya, M. Çolak, F. Terzi, A comprehensive review of fuzzy multi criteria

- decision making methodologies for energy policy making, *Energy Strateg. Rev.* 24 (2019) 207–228.
- [11] Y. Simsek, D. Watts, R. Escobar, Sustainability evaluation of concentrated solar power (CSP) projects under clean development mechanism (CDM) by using multi criteria decision method (MCDM), *Renew. Sustain. Energy Rev.* 93 (2018) 421–438.
 - [12] A. Kumar, B. Sah, A.R. Singh, Y. Deng, X. He, P. Kumar, et al., A review of multi criteria decision making (MCDM) towards sustainable renewable energy development, *Renew. Sustain. Energy Rev.* 69 (2017) 596–609.
 - [13] A. Mardani, E.K. Zavadskas, Z. Khalifah, N. Zakuan, A. Jusoh, K.M. Nor, et al., A review of multi-criteria decision-making applications to solve energy management problems: two decades from 1995 to 2015, *Renew. Sustain. Energy Rev.* 71 (2017) 216–256.
 - [14] S.D. Pohekar, M. Ramachandran, Application of multi-criteria decision making to sustainable energy planning - a review, *Renew. Sustain. Energy Rev.* 8 (2004) 365–381.
 - [15] V. Campos-Guzmán, M.S. García-Cascales, N. Espinosa, A. Urbina, Life Cycle Analysis with Multi-Criteria Decision Making: a review of approaches for the sustainability evaluation of renewable energy technologies, *Renew. Sustain. Energy Rev.* 104 (2019) 343–366.
 - [16] M. Tavana, F.J. Santos Arteaga, S. Mohammadi, M. Alimohammadi, A fuzzy multi-criteria spatial decision support system for solar farm location planning, *Energy Strateg. Rev.* 18 (2017) 93–105.
 - [17] A. Alami Merrouni, F. Elwali Elalaoui, A. Ghennoui, A. Mezhrab, A. Mezhrab, A GIS-AHP combination for the sites assessment of large-scale CSP plants with dry and wet cooling systems. Case study: eastern Morocco, *Sol. Energy* 166 (2018) 2–12.
 - [18] A. Asakereh, M. Soleymani, M.J. Sheikhdavoodi, A GIS-based Fuzzy-AHP method for the evaluation of solar farms locations: case study in Khuzestan province, Iran, *Sol. Energy* 155 (2017) 342–353.
 - [19] A. Maleki, F. Pourfayaz, H. Hafeznia, M.A. Rosen, A novel framework for optimal photovoltaic size and location in remote areas using a hybrid method: a case study of eastern Iran, *Energy Convers. Manag.* 153 (2017) 129–143.
 - [20] P. Aragonés-Beltrán, F. Chaparro-González, J.P. Pastor-Ferrando, F. Rodríguez-Pozo, An ANP-based approach for the selection of photovoltaic solar power plant investment projects, *Renew. Sustain. Energy Rev.* 14 (2010) 249–264.
 - [21] C. Perpiñá Castillo, F. Batista e Silva, C. Lavallo, An assessment of the regional potential for solar power generation in EU-28, *Energy Pol.* 88 (2016) 86–99.
 - [22] I. Potić, R. Golić, T. Joksimović, Analysis of insolation potential of Knjaževac Municipality (Serbia) using multi-criteria approach, *Renew. Sustain. Energy Rev.* 56 (2016) 235–245.
 - [23] M. Vafaeipour, S. Hashemkhani Zolfani, M.H. Morshed Varzandeh, A. Derakhti, M. Keshavarz Eshkalag, Assessment of regions priority for implementation of solar projects in Iran: new application of a hybrid multi-criteria decision making approach, *Energy Convers. Manag.* 86 (2014) 653–663.
 - [24] I. Gherboudj, H. Ghedira, Assessment of solar energy potential over the United Arab Emirates using remote sensing and weather forecast data, *Renew. Sustain. Energy Rev.* 55 (2016) 1210–1224.
 - [25] J.M. Sánchez-Lozano, M.S. García-Cascales, M.T. Lamata, Comparative TOPSIS-ELECTRE TRI methods for optimal sites for photovoltaic solar farms. Case study in Spain, *J. Clean. Prod.* 127 (2016) 387–398.
 - [26] T. Wanderer, S. Herle, Creating a spatial multi-criteria decision support system for energy related integrated environmental impact assessment, *Environ. Impact Assess. Rev.* 52 (2015) 2–8.
 - [27] Y. Wu, S. Geng, H. Zhang, M. Gao, Decision framework of solar thermal power plant site selection based on linguistic Choquet operator, *Appl. Energy* 136 (2014) 303–311.
 - [28] J. Arán Carrión, A. Espín Estrella, F. Aznar Dols, M. Zamorano Toro, M. Rodríguez, A. Ramos Rida, Environmental decision-support systems for evaluating the carrying capacity of land areas: optimal site selection for grid-connected photovoltaic power plants, *Renew. Sustain. Energy Rev.* 12 (2008) 2358–2380.
 - [29] D. Doljak, G. Stanojević, Evaluation of natural conditions for site selection of ground-mounted photovoltaic power plants in Serbia, *Energy* 127 (2017) 291–300.
 - [30] J.M. Sánchez-Lozano, M.S. García-Cascales, M.T. Lamata, Evaluation of suitable locations for the installation of solar thermoelectric power plants, *Comput. Ind. Eng.* 87 (2015) 343–355.
 - [31] J.M. Sánchez-Lozano, J. Teruel-Solano, P.L. Soto-Elvira, M. Socorro García-Cascales, Geographical Information Systems (GIS) and Multi-Criteria Decision Making (MCDM) methods for the evaluation of solar farms locations: case study in south-eastern Spain, *Renew. Sustain. Energy Rev.* 24 (2013) 544–556.
 - [32] A. Yushchenko, A. de Bono, B. Chatenoux, M.K. Patel, N. Ray, GIS-based assessment of photovoltaic (PV) and concentrated solar power (CSP) generation potential in West Africa, *Renew. Sustain. Energy Rev.* 81 (2018) 2088–2103.
 - [33] J.M. Sánchez-Lozano, C. Henggeler Antunes, M.S. García-Cascales, L.C. Dias, GIS-based photovoltaic solar farms site selection using ELECTRE-TRI: evaluating the case for Torre Pacheco, Murcia, Southeast of Spain, *Renew. Energy* 66 (2014) 478–494.
 - [34] M. Uyan, GIS-based solar farms site selection using analytic hierarchy process (AHP) in Karapinar region Konya/Turkey, *Renew. Sustain. Energy Rev.* 28 (2013) 11–17.
 - [35] S. Sindhu, V. Nehra, S. Luthra, Investigation of feasibility study of solar farms deployment using hybrid AHP-TOPSIS analysis: case study of India, *Renew. Sustain. Energy Rev.* 73 (2017) 496–511.
 - [36] A. Alami Merrouni, F. Elwali Elalaoui, A. Mezhrab, A. Mezhrab, A. Ghennoui, Large scale PV sites selection by combining GIS and Analytical Hierarchy Process. Case study: eastern Morocco, *Renew. Energy* 119 (2018) 863–873.
 - [37] L. Dawson, P. Schlyter, Less is more: strategic scale site suitability for concentrated solar thermal power in Western Australia, *Energy Pol.* 47 (2012) 91–101.
 - [38] S. Ozdemir, G. Sahin, Multi-criteria decision-making in the location selection for a solar PV power plant using AHP, *Meas. J. Int. Meas. Confed.* 129 (2018) 218–226.
 - [39] J.R.S. Doorga, S.D.D.V. Rughooputh, R. Boojhawon, Multi-criteria GIS-based modelling technique for identifying potential solar farm sites: a case study in Mauritius, *Renew. Energy* (2019) 1201–1219.
 - [40] M. Zoghi, A. Houshang Ehsani, M. Sadat, M. Javad Amiri, S. Karimi, Optimization solar site selection by fuzzy logic model and weighted linear combination method in arid and semi-arid region: a case study Isfahan-Iran, *Renew. Sustain. Energy Rev.* 68 (2017) 986–996.
 - [41] M. Azizkhani, A. Vakili, Y. Noorollahi, F. Naseri, Potential survey of photovoltaic power plants using Analytical Hierarchy Process (AHP) method in Iran, *Renew. Sustain. Energy Rev.* 75 (2017) 1198–1206.
 - [42] Y. Charabi, A. Gastli, PV site suitability analysis using GIS-based spatial fuzzy multi-criteria evaluation, *Renew. Energy* 36 (2011) 2554–2561.
 - [43] J. Liu, F. Xu, S. Lin, Site selection of photovoltaic power plants in a value chain based on grey cumulative prospect theory for sustainability: a case study in Northwest China, *J. Clean. Prod.* 148 (2017) 386–397.
 - [44] A. Aly, S.S. Jensen, A.B. Pedersen, Solar power potential of Tanzania: identifying CSP and PV hot spots through a GIS multicriteria decision making analysis, *Renew. Energy* 113 (2017) 159–175.
 - [45] H.Z. Al Garni, A. Awasthi, Solar PV power plant site selection using a GIS-AHP based approach with application in Saudi Arabia, *Appl. Energy* 206 (2017) 1225–1240.
 - [46] H. Fang, J. Li, W. Song, Sustainable site selection for photovoltaic power plant: an integrated approach based on prospect theory, *Energy Convers. Manag.* 174 (2018) 755–768.
 - [47] M. Tahri, M. Hakdaoui, M. Maanan, The evaluation of solar farm locations applying Geographic Information System and Multi-Criteria Decision-Making methods: case study in southern Morocco, *Renew. Sustain. Energy Rev.* 51 (2015) 1354–1362.
 - [48] J. Brewer, D.P. Ames, D. Solan, R. Lee, J. Carlisle, Using GIS analytics and social preference data to evaluate utility-scale solar power site suitability, *Renew. Energy* 81 (2015) 825–836.
 - [49] L.C. Rodman, R.K. Meentemeyer, A geographic analysis of wind turbine placement in Northern California, *Energy Pol.* 34 (2006) 2137–2149.
 - [50] D. Latinopoulos, K. Kechagia, A GIS-based multi-criteria evaluation for wind farm site selection. A regional scale application in Greece, *Renew. Energy* 78 (2015) 550–560.
 - [51] P.V. Gorsevski, S.C. Cathcart, G. Mirzaei, M.M. Jamali, X. Ye, E. Gomezdelcampo, A group-based spatial decision support system for wind farm site selection in Northwest Ohio, *Energy Pol.* 55 (2013) 374–385.
 - [52] T.R. Ayodele, A.S.O. Ogunjuyigbe, O. Odigie, J.L. Munda, A multi-criteria GIS based model for wind farm site selection using interval type-2 fuzzy analytic hierarchy process: the case study of Nigeria, *Appl. Energy* 228 (2018) 1853–1869.
 - [53] L. Givović, D. Pamučar, D. Božanić, S. Ljubojević, Application of the GIS-DANP-MABAC multi-criteria model for selecting the location of wind farms: a case study of Vojvodina, Serbia, *Renew. Energy* 103 (2017) 501–521.
 - [54] T. Parry, S. Baban, Developing and applying a GIS-assisted approach to locating wind farms in the UK, *Renew. Energy* 24 (2001) 59–71.
 - [55] L.I. Tegou, H. Polatidis, D.A. Haralambopoulos, Environmental management framework for wind farm siting: methodology and case study, *J. Environ. Manag.* 91 (2010) 2134–2147.
 - [56] M. Qolipour, A. Mostafaeipour, S. Shamsirband, O. Alavi, H. Goudarzi, D. Petković, Evaluation of wind power generation potential using a three hybrid approach for households in Ardebil Province, Iran, *Energy Convers. Manag.* 118 (2016) 295–305.
 - [57] N.Y. Aydin, E. Kentel, S. Duzgun, GIS-based environmental assessment of wind energy systems for spatial planning: a case study from Western Turkey, *Renew. Sustain. Energy Rev.* 14 (2010) 364–373.
 - [58] J.M. Sánchez-Lozano, M.S. García-Cascales, M.T. Lamata, GIS-based onshore wind farm site selection using Fuzzy Multi-Criteria Decision Making methods. Evaluating the case of Southeastern Spain, *Appl. Energy* 171 (2016) 86–102.
 - [59] M.A. Baseer, S. Rehman, J.P. Meyer, M.M. Alam, GIS-based site suitability analysis for wind farm development in Saudi Arabia, *Energy* 141 (2017) 1166–1176.
 - [60] J.M. Sánchez-Lozano, M.S. García-Cascales, M.T. Lamata, Identification and selection of potential sites for onshore wind farms development in region of Murcia, Spain, *Energy* 73 (2014) 311–324.
 - [61] A. Azadeh, S.F. Ghaderi, M.R. Nasrollahi, Location optimization of wind plants in Iran by an integrated hierarchical Data Envelopment Analysis, *Renew.*

- Energy 36 (2011) 1621–1631.
- [62] A. Azadeh, A. Rahimi-Golkhandan, M. Moghaddam, Location optimization of wind power generation-transmission systems under uncertainty using hierarchical fuzzy DEA: a case study, *Renew. Sustain. Energy Rev.* 30 (2014) 877–885.
- [63] A.H.I. Lee, H.H. Chen, H.Y. Kang, Multi-criteria decision making on strategic selection of wind farms, *Renew. Energy* 34 (2009) 120–126.
- [64] Y. Noorollahi, H. Yousefi, M. Mohammadi, Multi-criteria decision support system for wind farm site selection using GIS, *Sustain. Energy Technol. Assessments* 13 (2016) 38–50.
- [65] T. Kaya, C. Kahraman, Multicriteria renewable energy planning using an integrated fuzzy VIKOR & AHP methodology: the case of Istanbul, *Energy* 35 (2010) 2517–2527.
- [66] T.R. Ayodele, A.S.O. Ogunjuyigbe, O. Odigie, A.A. Jimoh, On the most suitable sites for wind farm development in Nigeria, *Data Br.* 19 (2018) 29–41.
- [67] J. Jangid, A.K. Bera, M. Joseph, V. Singh, T.P. Singh, B.K. Pradhan, et al., Potential zones identification for harvesting wind energy resources in desert region of India – a multi criteria evaluation approach using remote sensing and GIS, *Renew. Sustain. Energy Rev.* 65 (2016) 1–10.
- [68] M. Rezaei-Shouroki, A. Mostafaeipour, M. Qolipour, Prioritizing of wind farm locations for hydrogen production: a case study, *Int. J. Hydrogen Energy* 42 (2017) 9500–9510.
- [69] I.J. Ramírez-Rosado, E. García-Garrido, L.A. Fernández-Jiménez, P.J. Zorzano-Santamaría, C. Monteiro, V. Miranda, Promotion of new wind farms based on a decision support system, *Renew. Energy* 33 (2008) 558–566.
- [70] Y. Wu, S. Geng, H. Xu, H. Zhang, Study of decision framework of wind farm project plan selection under intuitionistic fuzzy set and fuzzy measure environment, *Energy Convers. Manag.* 87 (2014) 274–284.
- [71] G. Gamboa, G. Munda, The problem of windfarm location: a social multi-criteria evaluation framework, *Energy Pol.* 35 (2007) 1564–1583.
- [72] M. Eichhorn, P. Tafarte, D. Thrän, Towards energy landscapes – “Pathfinder for sustainable wind power locations, *Energy* 134 (2017) 611–621.
- [73] D. Mann, C. Lant, J. Schoof, Using map algebra to explain and project spatial patterns of wind energy development in Iowa, *Appl. Geogr.* 34 (2012) 219–229.
- [74] A. Simão, P.J. Densham, Muki, M. Haklay, Web-based GIS for collaborative planning and public participation: an application to the strategic planning of wind farm sites, *J. Environ. Manag.* 90 (2009) 2027–2040.
- [75] S. Al-Yahyai, Y. Charabi, A. Gastil, A. Al-Badi, Wind farm land suitability indexing using multi-criteria analysis, *Renew. Energy* 44 (2012) 80–87.
- [76] T. Höfer, Y. Sunak, H. Siddique, R. Madlener, Wind farm siting using a spatial Analytic Hierarchy Process approach: a case study of the Städteregion Aachen, *Appl. Energy* 163 (2016) 222–243.
- [77] G. Villacreses, G. Gaona, J. Martínez-Gómez, D.J. Jijón, Wind farms suitability location using geographical information system (GIS), based on multi-criteria decision making (MCDM) methods: the case of continental Ecuador, *Renew. Energy* 109 (2017) 275–286.
- [78] J.R. San Cristóbal, A goal programming model for the optimal mix and location of renewable energy plants in the north of Spain, *Renew. Sustain. Energy Rev.* 16 (2012) 4461–4464.
- [79] N.Y. Aydin, E. Kentel, H. Sebnem Duzgun, GIS-based site selection methodology for hybrid renewable energy systems: a case study from western Turkey, *Energy Convers. Manag.* 70 (2013) 90–106.
- [80] M. Rezaei, A. Mostafaeipour, M. Qolipour, R. Tavakkoli-Moghaddam, Investigation of the optimal location design of a hybrid wind-solar plant: a case study, *Int. J. Hydrogen Energy* 43 (2018) 100–114.
- [81] D. Jun, F. Tian-Tian, Y. Yi-Sheng, M. Yu, Macro-site selection of wind/solar hybrid power station based on ELECTRE-II, *Renew. Sustain. Energy Rev.* 35 (2014) 194–204.
- [82] W. Yun-Na, Y. Yi-Sheng, F. Tian-Tian, K. Li-Na, L. Wei, F. Luo-Jie, Macro-site selection of wind/solar hybrid power station based on Ideal Matter-Element Model, *Int. J. Electr. Power Energy Syst.* 50 (2013) 76–84.
- [83] W. Yunna, S. Geng, Multi-criteria decision making on selection of solar-wind hybrid power station location: a case of China, *Energy Convers. Manag.* 81 (2014) 527–533.
- [84] J.R. Janke, Multicriteria GIS modeling of wind and solar farms in Colorado, *Renew. Energy* 35 (2010) 2228–2234.
- [85] J.J.W. Watson, M.D. Hudson, Regional Scale wind farm and solar farm suitability assessment using GIS-assisted multi-criteria evaluation, *Landsc. Urban Plann.* 138 (2015) 20–31.
- [86] H.H. Chen, H.Y. Kang, A.H.I. Lee, Strategic selection of suitable projects for hybrid solar-wind power generation systems, *Renew. Sustain. Energy Rev.* 14 (2010) 413–421.
- [87] B. Wu, T.L. Yip, L. Xie, Y. Wang, A fuzzy-MADM based approach for site selection of offshore wind farm in busy waterways in China, *Ocean Eng.* 168 (2018) 121–132.
- [88] A. Fetanat, E. Khorasaninejad, A novel hybrid MCDM approach for offshore wind farm site selection: a case study of Iran, *Ocean Coast Manag.* 109 (2015) 17–28.
- [89] A.D. Mekonnen, P.V. Gorsevski, A web-based participatory GIS (PGIS) for offshore wind farm suitability within Lake Erie, Ohio, *Renew. Sustain. Energy Rev.* 41 (2015) 162–177.
- [90] Z. Xue-yang, W. Xiao-kang, Y. Su-min, W. Jian-qiang, W. Tie-li, Location selection of offshore wind power station by consensus decision framework using picture fuzzy modelling, *J. Clean. Prod.* 202 (2018) 980–992.
- [91] M. Mahdy, A.B.S. Bahaj, Multi criteria decision analysis for offshore wind energy potential in Egypt, *Renew. Energy* 118 (2018) 278–289.
- [92] A. Chaouachi, C.F. Covrig, M. Ardelean, Multi-criteria selection of offshore wind farms: case study for the Baltic States, *Energy Pol.* 103 (2017) 179–192.
- [93] Y. Wu, J. Zhang, J. Yuan, S. Geng, H. Zhang, Study of decision framework of offshore wind power station site selection based on ELECTRE-III under intuitionistic fuzzy environment: a case of China, *Energy Convers. Manag.* 113 (2016) 66–81.
- [94] M.M. Abaei, E. Arzaghi, R. Abbassi, V. Garaniya, I. Penesis, Developing a novel risk-based methodology for multi-criteria decision making in marine renewable energy applications, *Renew. Energy* 102 (2017) 341–348.
- [95] S. Ghosh, T. Chakraborty, S. Saha, M. Majumder, M. Pal, Development of the location suitability index for wave energy production by ANN and MCDM techniques, *Renew. Sustain. Energy Rev.* 59 (2016) 1017–1028.
- [96] A. Nobre, M. Pacheco, R. Jorge, M.F.P. Lopes, L.M.C. Gato, Geo-spatial multi-criteria analysis for wave energy conversion system deployment, *Renew. Energy* 34 (2009) 97–111.
- [97] A.H.I. Lee, H.H. Chen, H.Y. Kang, A conceptual model for prioritizing dam sites for tidal energy sources, *Ocean Eng.* 137 (2017) 38–47.
- [98] Y. Wu, C. Xu, Y. Ke, K. Chen, H. Xu, Multi-criteria decision-making on assessment of proposed tidal barrage schemes in terms of environmental impacts, *Mar. Pollut. Bull.* 125 (2017) 271–281.
- [99] M. Vasileiou, E. Loukogeorgaki, D.G. Vagiona, GIS-based multi-criteria decision analysis for site selection of hybrid offshore wind and wave energy systems in Greece, *Renew. Sustain. Energy Rev.* 73 (2017) 745–757.
- [100] L. Cradden, C. Kalogeri, I.M. Barrios, G. Galanis, D. Ingram, G. Kallos, Multi-criteria site selection for offshore renewable energy platforms, *Renew. Energy* 87 (2016) 791–806.
- [101] A. Moldavska, T. Welo, The concept of sustainable manufacturing and its definitions: a content-analysis based literature review, *J. Clean. Prod.* 166 (2017) 744–755.
- [102] Özyurt Ö, H. Özyurt, Learning style based individualized adaptive e-learning environments: content analysis of the articles published from 2005 to 2014, *Comput. Hum. Behav.* 52 (2015) 349–358.
- [103] Y.J. Guo, System Synthetical Evaluation Theory, Methods and Application, Science Press, Beijing, 2012.
- [104] L. Zadeh, Fuzzy sets, *Inf. Contr.* 8 (1965) 338–353.
- [105] L. Zadeh, The concept of a linguistic variable and its application to approximate reasoning—II, *Inf. Sci.* 8 (1975) 301–357.
- [106] R.M. Dawes, B. Corrigan, Linear models in decision making, *Psychol. Bull.* 81 (1974) 95–106.
- [107] T.L. Saaty, The Analytic Hierarchy Process, McGraw-Hill, New York, 1980.
- [108] T.L. Saaty, The Analytic Network Process: Decision-Making with Dependence and Feedback, RWS Publications, Pittsburgh, 1996.
- [109] R.R. Yager, Quantifier guided aggregation using OWA operators, *Int. J. Intell. Syst.* 11 (1996) 49–73.
- [110] G. Munda, Social Multi-Criteria Evaluation for a Sustainable Economy, Springer, Berlin, 2008.
- [111] J. Figueira, B. Roy, Determining the weights of criteria in the ELECTRE type methods with a revised Simos’ procedure, *Eur. J. Oper. Res.* 139 (2002) 317–326.
- [112] C.L. Hwang, K. Yoon, Multiple Attribute Decision Methods and Applications, Springer, Berlin Heidelberg, 1981.
- [113] R. Benayoun, B. Roy, B. Sussman, ELECTRE: une méthode pour guider le choix en présence de points de vue multiples, *Note De Trav* 49 (1966).
- [114] R.R. Yager, On ordered weighted averaged aggregation operator in multi-criteria decision making, *IEEE Trans. Syst. Cybernetic* 18 (1988) 183–190.
- [115] A. Charnes, W.W. Cooper, E. Rhodes, Measuring the efficiency of decision making units, *Eur. J. Oper. Res.* 2 (1978) 429–444.
- [116] S. Opricovic, Multicriteria optimization of civil engineering systems, *Fac Civ. Eng. Belgrade* 2 (1998) 5–21.
- [117] D. Pamučar, G. Čirović, The selection of transport and handling resources in logistics centers using Multi-Attributive Border Approximation area Comparison (MABAC), *Expert Syst. Appl.* 42 (2015) 3016–3028.